Critical mean out-degrees of directed, undirected and bidirectional lattice neighbour graphs

Benedikt Jahnel*

in conversation with Claude Opus 5.5 (Anthropic)

1 November 2026

Abstract

In lattice neighbour graphs on $\mathbb{Z}^d$ every vertex independently chooses a random set of nearest neighbours whose mean size $\beta \in [0, 2d]$ varies continuously. From these choices one obtains a directed graph and two undirected graphs, in which an edge is open if one, respectively both, of its endpoints choose it. We study the three critical mean out-degrees $\beta_c^{\mathrm{U}}(d) \le \beta_c^{\mathrm{D}}(d) \le \beta_c^{\mathrm{B}}(d)$. Simulations in dimensions $2 \le d \le 10$ show that they behave very differently: $\beta_c^{\mathrm{B}}(d)$ grows like $\sqrt{2d}$, $\beta_c^{\mathrm{D}}(d)$ decreases like $1 + 1/(2d)$, and $\beta_c^{\mathrm{U}}(d)$ approaches one like $1 + 1/(8d^2)$. We prove that $\beta_c^{\mathrm{B}}(d) = \sqrt{2d} + 1/2 + O(d^{-1/2})$ and $\beta_c^{\mathrm{D}}(d) = 1 + 1/(2d) + O(d^{-2})$, and that $1+(1+o(1))/(8d^2) \le \beta_c^{\mathrm{U}}(d) \le 1+1/(2d)+O(d^{-2})$. The lower bound $\beta_c^{\mathrm{B}}(d) \ge \sqrt{2d}+1/2$ holds in every dimension $d \ge 2$. In addition, we give explicit two-sided bounds in all dimensions, which for $d \ge 11$ determine $\beta_c^{\mathrm{B}}(d)$ up to about 0.03, and we prove $\beta_c^{\mathrm{B}}(2) < 3$. A column projection gives upper bounds on $\beta_c^{\mathrm{D}}(d)$ for all $d \ge 3$ and shows $\beta_c^{\mathrm{D}}(d) < \beta_c^{\mathrm{D}}(2)$ for $d \ge 8$. The main tool is a sharp local comparison between the mixture of uniform k- and $(k+1)$-subsets and a product law, which holds for both undirected rules.

1 Introduction

In the lattice k-neighbour graph every vertex of $\mathbb{Z}^d$ independently chooses k of its $2d$ nearest neighbours uniformly at random [JKLT25]. The choices define three graphs: the directed graph with an arrow from every vertex to each chosen neighbour, the undirected graph in which an edge is open if at least one endpoint chooses it, and the bidirectional graph in which an edge is open if both endpoints choose it. The integer k is a coarse parameter, and [CHJK26, BJK+25] introduced a continuous interpolation in which every vertex chooses $\lfloor\beta\rfloor$ neighbours and, with probability $\beta - \lfloor\beta\rfloor$, one more. The mean out-degree β then varies in $[0, 2d]$, and each of the three graphs has a critical value $\beta_c^{\mathrm{U}}(d) \le \beta_c^{\mathrm{D}}(d) \le \beta_c^{\mathrm{B}}(d)$.

This note gives a joint picture of the three critical curves. Simulations (Figure 2) show three different behaviours. The bidirectional threshold grows like $\sqrt{2d}$, because two neighbours have to agree on an edge. The directed threshold decreases towards one like $1 + 1/(2d)$. The undirected threshold is already close to one in dimension three, and the simulations suggest $\beta_c^{\mathrm{U}}(d) = 1 + (1 + o(1))/(8d^2)$ (Conjecture 8.1). We then collect what can be proved. Our main results are the following.

*Institut für Mathematische Stochastik, Technische Universität Braunschweig, Germany, and Weierstrass Institute for Applied Analysis and Stochastics, Berlin, Germany. benedikt.jahnel@tu-braunschweig.de. ORCID: 0000-0002-4212-0065.

- A sharp local comparison of both undirected graphs with Bernoulli bond percolation for all $\beta \in [0, 2d]$ (Theorem 2.2). For the bidirectional graph the comparison parameter is the probability that a vertex keeps two given edges, which is optimal.
- The asymptotics $\beta_c^{\mathrm{B}}(d) = \sqrt{2d} + 1/2 + O(d^{-1/2})$ and $\beta_c^{\mathrm{D}}(d) = 1 + 1/(2d) + O(d^{-2})$, together with $1 + (1 + o(1))/(8d^2) \le \beta_c^{\mathrm{U}}(d) \le 1 + 1/(2d) + O(d^{-2})$ (Theorem 2.5). The lower bound $\beta_c^{\mathrm{B}}(d) \ge \sqrt{2d} + 1/2$ holds in all dimensions $d \ge 2$.
- Explicit two-sided bounds in every dimension (Table 2, Figure 3), which are close to sharp for the bidirectional graph in dimensions $d \ge 11$, and the strict inequality $\beta_c^{\mathrm{B}}(2) < 3$ (Theorem 2.6).
- Explicit upper bounds on $\beta_c^{\mathrm{D}}(d)$ in every dimension $d \ge 3$, for instance $\beta_c^{\mathrm{D}}(3) \le 1.834$, $\beta_c^{\mathrm{D}}(10) \le 1.422$ and $\beta_c^{\mathrm{D}}(d) \le 1.457$ for all $d \ge 9$, from a column projection that reduces the directed graph to the plane (Theorem 2.7); for $\beta = 2$ the same reduction was used in [LXZ26]. With a comparison principle between dimensions we also show $\beta_c^{\mathrm{D}}(7) \le \beta_c^{\mathrm{D}}(2)$ and $\beta_c^{\mathrm{D}}(d) < \beta_c^{\mathrm{D}}(2)$ for $d \ge 8$ (Theorem 2.9), the first comparisons between dimensions for this model.

Previously, $\lim_{d\to\infty} \beta_c^{\mathrm{B}}(d)/\sqrt{2d} = 1$ [BJK+25, Proposition 3.8] and $\lim_{d\to\infty} \beta_c^{\mathrm{D}}(d) = 1$, with the upper bound from [BJK+25, Corollary 3.7] and the lower bound from [CHJK26, Lemma 2.2], were known.

Section 2 introduces the models, presents the simulations and states the results. Section 3 contains the local comparison, Section 4 the bounds for absence of percolation, and Section 5 the proofs of the asymptotic results, Section 6 the column projection for the directed graph, and Section 7 the enhancement argument for the planar bidirectional graph. Section 8 discusses open problems.

2 Setting, simulations and main results

2.1 The models

For $d \in \mathbb{N}$ let $\mathcal{D}_d := \{\pm e_1, \dots, \pm e_d\}$ and $m := 2d$. For $0 \le k \le m$ let $\upsilon_{d,k}$ be the uniform distribution on the k-subsets of $\mathcal{D}_d$. For $\beta \in [0, m]$ write $\beta = k + \varepsilon$ with $k := \lfloor \beta \rfloor$ and $\varepsilon \in [0, 1)$ (for $\beta = m$ we set $k = m$, $\varepsilon = 0$), and let

$$Q_{d,\beta} := (1 - \varepsilon)\, \upsilon_{d,k} + \varepsilon\, \upsilon_{d,k+1}\,.$$

Every vertex $x \in \mathbb{Z}^d$ independently draws a set $N_x \subseteq \mathcal{D}_d$ with law $Q_{d,\beta}$; the mean of $\#N_x$ is β. We say that x *keeps* the edge $\{x, x + u\}$ if $u \in N_x$. The three graphs are the following.

- *Directed* (D): a directed edge from x to $x + u$ for every $u \in N_x$.
- *Undirected* (U): the edge $\{x, x + u\}$ is open if $u \in N_x$ or $-u \in N_{x+u}$.
- *Bidirectional* (B): the edge $\{x, x + u\}$ is open if $u \in N_x$ and $-u \in N_{x+u}$.

Figure 1 shows the three graphs for one configuration.

For $\star \in \{\mathrm{B}, \mathrm{D}, \mathrm{U}\}$ let $\theta_d^\star(\beta)$ be the probability that the origin o has an infinite open cluster, respectively an infinite out-cluster in the directed case. In the coupling in which every vertex draws a uniform order of $\mathcal{D}_d$ and a uniform variable $U_x \in [0, 1)$ and keeps the direction in position $i \in \{0, \dots, m - 1\}$ if $\beta \ge i + U_x$, the sets N_x increase in β. Hence $\theta_d^\star$ is non-decreasing, and we

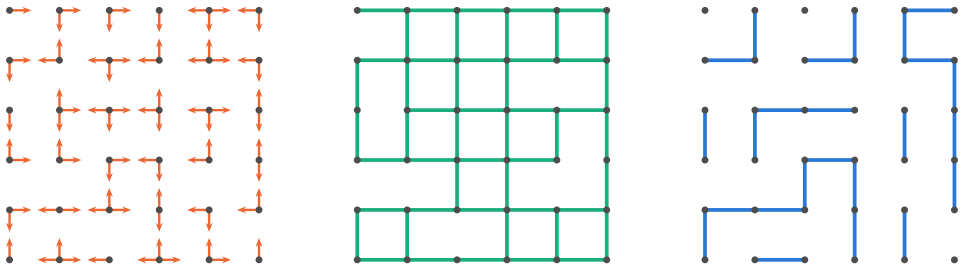


Figure 1: One configuration of the keep sets on a 6×6 patch of $\mathbb{Z}^2$ with mean out-degree $\beta = 2.5$: every vertex keeps two uniformly chosen directions and, with probability $1/2$, a third one. Left: the directed graph, with an arrow along half of the edge for every kept direction, so that an edge carries two arrows if and only if it is open in the bidirectional graph. Middle: the undirected graph, in which an edge is open if at least one endpoint keeps it. Right: the bidirectional graph, in which an edge is open if both endpoints keep it.

define

$$\beta_c^\star(d) := \inf\{\beta \in [0, 2d] \colon \theta_d^\star(\beta) > 0\}\,.$$

Since an open bidirectional path is a directed path and the edges of a directed path are open in the undirected graph,

$$\beta_c^{\mathrm{U}}(d) \le \beta_c^{\mathrm{D}}(d) \le \beta_c^{\mathrm{B}}(d)\,. \tag{2.1}$$

In the notation of [BJK+25], $\beta_c^\star(d) = 2d\, p_c^\star(d)$. We write $p_c(d)$ for the critical value of Bernoulli bond percolation on $\mathbb{Z}^d$ and $c(d)$ for the connective constant of $\mathbb{Z}^d$. A quantity that appears throughout is

$$\varphi(\beta) := k(k-1+2\varepsilon) = (1-\varepsilon)\,k(k-1) + \varepsilon\,(k+1)k\,, \qquad \beta = k + \varepsilon\,.$$

The probability that a vertex keeps two given edges equals $\varphi(\beta)/(m(m-1))$, whereas a given edge is kept by one endpoint with probability β/m and is open in the bidirectional graph with probability $(\beta/m)^2$. The function φ is continuous, increasing on $[1, m]$ and linear with slope $2k$ on $[k, k+1]$.

2.2 Simulations

We estimated $\beta_c^{\mathrm{B}}(d)$ and $\beta_c^{\mathrm{U}}(d)$ for $2 \le d \le 10$ with two methods, both based on the monotone coupling above.

- *Spanning thresholds* ($d \le 4$). For every sample in the box $\{0, \dots, L-1\}^d$ the edges are added in increasing order of the value of β at which they become open, and we record the value at which the faces $\{x_1 = 0\}$ and $\{x_1 = L-1\}$ become connected. The empirical distribution functions of this threshold for L and $2L$ cross close to the critical value. In $d = 2$ we use $L \in \{512, 1024\}$ with 1000 and 600 samples; the crossings for $L \in \{64, \dots, 1024\}$ agree within 0.002.

- *Mean cluster size* ($d \ge 3$). On the infinite lattice, with keep sets generated when needed, we estimate the mean size $\chi(\beta)$ of the cluster of the origin from 3000 samples for a sequence of subcritical β with $\chi(\beta)$ up to several hundred or a few thousand, and extrapolate $\chi^{-1/\gamma}$

linearly to zero. We use the exponent γ of Bernoulli percolation, $\gamma = 1.80, 1.44, 1.18$ for $d = 3, 4, 5$ and $\gamma = 1$ for $d \geq 6$.

In $d \in \{3, 4\}$ both methods agree within 0.0015. For $d \geq 5$ the boxes that can be handled are too small for the first method. For the directed graph we use the values of [CHJK26, Table 1] for $d \leq 4$, and for $d \geq 5$ the mean size of the out-cluster of the origin with the same extrapolation. The critical exponent of the out-cluster is not known, and in $d \in \{3, 4\}$ the extrapolation with the exponents of Bernoulli percolation gives 1.40 and 1.22 instead of 1.38 and 1.20, so that we expect an uncertainty of about 0.02 for the directed values. Table 1 and Figure 2 show the results.

d	2	3	4	5	6	7	8	9	10
$\beta_c^{\mathrm{B}}(d)$	2.840	3.162	3.443	3.736	4.036	4.280	4.522	4.771	5.017
$\beta_c^{\mathrm{D}}(d)$	1.84	1.38	1.20	1.146	1.109	1.088	1.074	1.064	1.057
$\beta_c^{\mathrm{U}}(d)$	1.249	1.056	1.019	1.0093	1.0052	1.0034	1.0025	1.0019	1.0015
$\varphi^{-1}(2d)$	2.500	3.000	3.333	3.667	4.000	4.250	4.500	4.750	5.000
$1 + 1/(2d)$	1.250	1.167	1.125	1.100	1.083	1.071	1.063	1.056	1.050
$1 + 1/(8d^2)$	1.0313	1.0139	1.0078	1.0050	1.0035	1.0026	1.0020	1.0015	1.0013

Table 1: Simulated critical mean out-degrees. For B and U: $d = 2$ from the crossing of the spanning-threshold distributions for $L \in \{512, 1024\}$, $d \geq 3$ from the extrapolation of the mean cluster size; the uncertainty is about ± 0.005 for B and ± 0.001 for U; for $d \in \{9, 10\}$ only the points with $\beta \geq 1$ are used for U. For D: [CHJK26, Table 1] for $d \leq 4$, and the out-cluster extrapolation for $d \geq 5$, with an uncertainty of about 0.02. The last three rows are the rigorous lower bound for β_c^{B} and the leading terms for β_c^{D} and β_c^{U} from Theorem 2.5 and Conjecture 8.1.

The three curves behave differently. The bidirectional threshold lies only slightly above $\varphi^{-1}(2d) \geq \sqrt{2d} + 1/2$, which is a rigorous lower bound (Theorem 2.5); the difference is 0.036, 0.022 and 0.017 for $d = 6, 8, 10$. The directed threshold is close to $1 + 1/(2d)$ already in moderate dimensions. The undirected threshold is much closer to one than the directed one, and $8d^2(\beta_c^{\mathrm{U}}(d) - 1)$ takes the values 1.9, 1.5, 1.3, 1.3, 1.2 and 1.2 for $d = 5, \ldots, 10$, which suggests that the rigorous lower bound $1 + (1 + o(1))/(8d^2)$ is sharp.

2.3 Main results

The first result compares the three graphs with Bernoulli bond percolation. For $n \in \mathbb{N}$ let $B_n := \{x \in \mathbb{Z}^d \colon \|x\|_\infty \leq n\}$, $\partial B_n := \{x \colon \|x\|_\infty = n\}$, and let $\{o \longleftrightarrow \partial B_n\}$ be the event that o is connected to ∂B_n by an open path in B_n (a directed path in the directed graph). We write $\mathbb{P}^{\star}_{d,\beta}$ for the law of graph $\star$ and $\mathbb{P}^{\mathrm{bond}}_p$ for Bernoulli bond percolation with parameter p on $\mathbb{Z}^d$, and $\theta^{\mathrm{bond}}_d(p)$ for the probability that the open cluster of o is infinite under $\mathbb{P}^{\mathrm{bond}}_p$.

Theorem 2.2. *Let $d \in \mathbb{N}$, $\beta \in [0, 2d]$ and $q := \sqrt{\varphi(\beta)/(2d(2d-1))}$. Then for all $n \in \mathbb{N}$,*

$$\mathbb{P}^{\mathrm{B}}_{d,\beta}(o \longleftrightarrow \partial B_n) \geq \mathbb{P}^{\mathrm{bond}}_{q^2}(o \longleftrightarrow \partial B_n)\,, \qquad \mathbb{P}^{\mathrm{U}}_{d,\beta}(o \longleftrightarrow \partial B_n) \geq \mathbb{P}^{\mathrm{bond}}_{1-(1-q)^2}(o \longleftrightarrow \partial B_n)\,.$$

Moreover, $\mathbb{P}^{\mathrm{D}}_{d,\beta}(o \longleftrightarrow \partial B_n) \geq \mathbb{P}^{\mathrm{bond}}_{\beta/(2d)}(o \longleftrightarrow \partial B_n)$ [BJK+25, Corollary 3.3, Theorem 2.1 and Lemma 3.5].

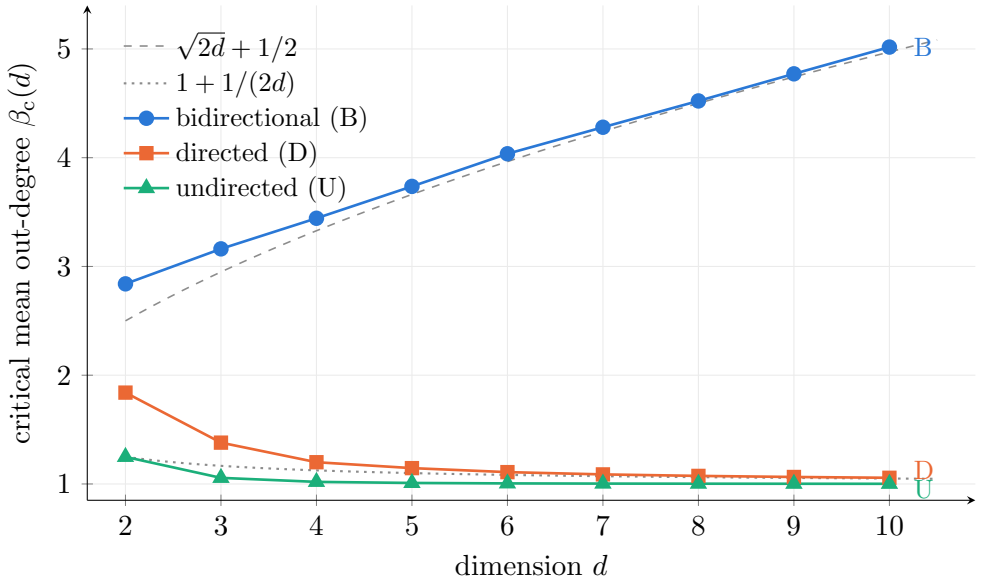


Figure 2: Simulated critical mean out-degrees of the bidirectional (circles), directed (squares) and undirected (triangles) lattice neighbour graphs, see Table 1. The dashed and dotted curves are the leading asymptotics of β_c^{B} and β_c^{D} from Theorem 2.5. The uncertainties are smaller than the markers, except for D with $d \geq 5$ (about 0.02).

The parameter q is optimal: for $|A| = |B| = 1$ equality holds in the local inequality behind Theorem 2.2, see Lemma 3.4. Moreover, $q^2 \leq (\beta/m)^2$, the probability that an edge is open in the bidirectional graph, by (3.6). The second result gives absence of percolation. For the undirected graph let

$$\bar{\lambda}_d(\beta) := \frac{\beta}{m} - \frac{\kappa}{2} + \sqrt{\kappa}\,, \qquad \kappa := \frac{\varphi(\beta)}{m(m-1)}\,.$$

Theorem 2.3. *Let $d \geq 2$ and $\beta \in [0, 2d]$. Then $\theta_d^\star(\beta) = 0$ in each of the following cases.*

(B) $\star = \mathrm{B}$ *and* $c(d)\,\varphi(\beta) < 2d(2d-1)$, *in particular if* $\varphi(\beta) \leq 2d$.

(D) $\star = \mathrm{D}$ *and* $c(d)\,\beta < 2d$, *in particular if* $\beta \leq 2d/(2d-1)$.

(U) $\star = \mathrm{U}$ *and* $\beta \leq 1$ *or* $c(d)\,\bar{\lambda}_d(\beta) < 1$.

Part (D) is [CHJK26, Lemma 2.2], and part (B) extends [JKLT25, Lemma 2.6] to non-integer β. Combined with the known upper bounds on $p_c(d)$ and $c(d)$, Theorems 2.2 and 2.3 give explicit bounds in every dimension.

Theorem 2.4. *For every $d \geq 2$,*

$$\varphi^{-1}\Big(\frac{2d(2d-1)}{c(d)}\Big) \leq \beta_c^{\mathrm{B}}(d) \leq \varphi^{-1}\big(2d(2d-1)p_c(d)\big)\,, \qquad \frac{2d}{c(d)} \leq \beta_c^{\mathrm{D}}(d) \leq 2d\,p_c(d)\,,$$

and $\beta_c^{\mathrm{U}}(d) \leq \min\{\beta_c^{\mathrm{D}}(d), \beta_*\}$, where β_* solves $1-(1-q)^2 = p_c(d)$ with $q^2 = \varphi(\beta_*)/(2d(2d-1))$. With the bounds on $p_c(d)$ and $c(d)$ of [GPS26, FvdH17, PT00], this gives the values in Table 2.

The bounds determine the asymptotic behaviour of β_c^{B} and β_c^{D}.

d	$\beta_c^B(d)$		$\beta_c^D(d)$		$\beta_c^U(d)$	
	lower	upper	lower	upper	lower	upper
2	2.6197	< 3	1.4929	< 2	1.0738	1.5148
3	3.0551	3.7365	1.2661	1.8340	1.0252	1.5536
4	3.3717	4.4513	1.1757	1.7110	1.0119	1.6362
5	3.6929	5.0550	1.1286	1.6210	1.0067	1.6210
6	4.0000	5.5358	1.0909	1.5530	1.0035	1.5530
7	4.2500	6.0498	1.0769	1.5120	1.0025	1.5120
8	4.5000	6.4703	1.0666	1.4770	1.0019	1.4770
9	4.7500	6.8804	1.0588	1.4470	1.0015	1.4470
10	5.0000	7.2708	1.0526	1.4220	1.0012	1.4220
11	5.2000	5.2288	1.0476	1.0614	1.0010	1.0614
12	5.4000	5.4206	1.0434	1.0525	1.0008	1.0525
15	6.0000	6.0098	1.0344	1.0386	1.0005	1.0386
20	6.8333	6.8393	1.0256	1.0275	1.0003	1.0275

Table 2: Rigorous bounds from Theorems 2.3, 2.4, 2.6 and 2.7, rounded outwards. For $d = 2$ the upper bounds are Theorem 2.6 and [CHJK26, Theorem 2.3]. For $d \geq 6$ the lower bounds use only $c(d) < 2d - 1$.

Theorem 2.5. *For all $d \geq 2$,*

$$\sqrt{2d} + \frac{1}{2} \leq \varphi^{-1}(2d) \leq \beta_c^B(d)\,, \qquad 1 + \frac{1}{2d-1} \leq \beta_c^D(d)\,,$$
$$1 + \Bigl(\frac{\sqrt{2+2/d} - \sqrt{2}}{2}\Bigr)^2 \leq \beta_c^U(d) \leq \beta_c^D(d)\,.$$

Moreover, as $d \to \infty$,

$$\beta_c^B(d) = \varphi^{-1}(2d) + O(d^{-3/2}) = \sqrt{2d} + \frac{1}{2} + O(d^{-1/2})\,, \qquad \beta_c^D(d) = 1 + \frac{1}{2d} + O(d^{-2})\,,$$

and

$$1 + \frac{1 + O(d^{-1})}{8d^2} \leq \beta_c^U(d) \leq 1 + \frac{1}{2d} + O(d^{-2})\,.$$

In particular, $\beta_c^D(d) < \beta_c^B(d)$ for all $d \geq 2$, since $\beta_c^D(d) \leq 2$ by [CHJK26, Theorem 2.3] and [LXZ26, Theorem 1.1], while $\beta_c^B(d) \geq \varphi^{-1}(4) = 5/2$. In the plane, the comparison of Theorem 2.2 is critical for the bidirectional graph at $\beta = 3$, since $\varphi(3)/12 = 1/2 = p_c(2)$. An enhancement argument in the spirit of [AG91], carried out in Section 7, gives a strict improvement.

Theorem 2.6. $2.6197 \leq \beta_c^B(2) < 3$.

For the directed graph, Theorem 2.2 gives useful upper bounds only in high dimensions, since $2d\,p_c(d) \geq 2$ for $3 \leq d \leq 10$ with the available bounds on $p_c(d)$. A column projection, which reduces the directed model on $\mathbb{Z}^d$ to a directed model on $\mathbb{Z}^2$ and compares the latter with Bernoulli bond percolation, gives explicit bounds in all dimensions $d \geq 3$.

Theorem 2.7. *For $3 \leq d \leq 30$ we have $\beta_c^D(d) \leq \bar{\beta}_d$ with the values $\bar{\beta}_d$ of Table 3, and $\beta_c^D(d) \leq 1.457$ for all $d \geq 31$. In particular,*

$$\sup_{d \geq 3} \beta_c^D(d) \leq 1.834\,, \qquad \sup_{d \geq 8} \beta_c^D(d) \leq 1.477\,, \qquad \sup_{d \geq 9} \beta_c^D(d) \leq 1.457\,.$$

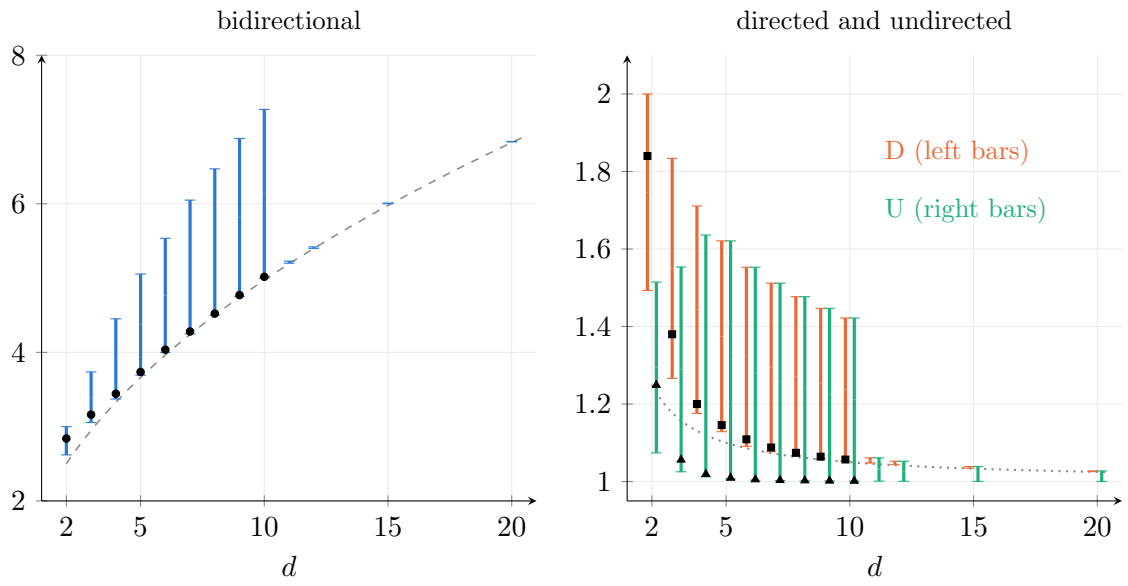


Figure 3: Rigorous bounds of Table 2 (vertical bars) and simulated values (black markers) for $d \in \{2, \dots, 12, 15, 20\}$. Left: bidirectional graph with $\sqrt{2d} + 1/2$ (dashed). Right: directed (left bars, squares) and undirected (right bars, triangles) graph with $1 + 1/(2d)$ (dotted). For $d \geq 11$ the bounds use the computer-assisted estimates on $p_c(d)$ of [FvdH17].

d	3	4	5	6	7	8	9	10	11	12
$\bar{\beta}_d$	1.834	1.711	1.621	1.553	1.512	1.477	1.447	1.422	1.405	1.389
d	13	14	15	16	17	18	19	20	21	22
$\bar{\beta}_d$	1.375	1.362	1.35	1.339	1.328	1.319	1.31	1.301	1.293	1.286
d	23	24	25	26	27	28	29	30	$\geq$31	
$\bar{\beta}_d$	1.279	1.272	1.266	1.26	1.255	1.249	1.244	1.241	1.457	

Table 3: Rigorous upper bounds $\bar{\beta}_d$ on $\beta_c^{\mathrm{D}}(d)$ from Theorem 2.7. For $d \in \{11, \dots, 18\} \cup \{20, \dots, 24\}$ the bounds $2d\, p_c(d)$ from Theorem 2.2, [FvdH17, Theorem 1.2] and $p_c(d) \leq p_c(15)$, $p_c(d) \leq p_c(20)$ are better, see Table 2 for some of them.

The bound for $d \geq 31$ is not obtained by a computation in each dimension. It rests on an algebraic inequality between consecutive dimensions, valid for all mean out-degrees $\beta \in [1.45, 2]$ (Lemma 6.13), which makes a closure-weakened version of the projection bound non-increasing in d. The same method gives the first comparison between two dimensions for the directed graph. By Theorem 2.3(D) and $c(2) \leq 2.679192495$ [PT00] we have

$$\beta_c^{\mathrm{D}}(2) \geq 4/2.679192495 > 1.4929\,. \tag{2.8}$$

Theorem 2.9. *We have $\beta_c^{\mathrm{D}}(7) \leq \beta_c^{\mathrm{D}}(2)$ and $\beta_c^{\mathrm{D}}(d) < \beta_c^{\mathrm{D}}(2)$ for all $d \geq 8$.*

For $d \geq 8$ this follows from Theorem 2.7 and (2.8). The case $d = 7$ uses that a five-fold projection of the 7-dimensional model dominates the planar model locally, for all mean out-degrees between the lower bound (2.8) and $\bar{\beta}_7$, see the comparison principle in Proposition 6.11.

2.4 Implications for integer k

For integer $\beta = k$ the model is the lattice k-neighbour graph of [JKLT25]. Table 4 updates [BJK+25, Table 1] with what Theorems 2.2, 2.3 and 2.6 give for the bidirectional graph. In $d \in \{2,3\} \cup \{11,\dots,17\} \cup \{20,\dots,23\}$ the integer threshold is determined, and it equals the smallest k with $k(k-1) > 2d$. For $2 \le d \le 10$ this is also the smallest integer above the simulated value of $\beta_c^{\mathrm{B}}(d)$. For the directed and the undirected graph the integer threshold is 2 in every dimension $d \ge 2$, so that the continuous parameter is needed to separate the three graphs.

d \ k	1	2	3	4	5	6	≥ 7
1	no/no/no	yes/yes/yes	–	–	–	–	–
2	no/no/no	no/yes/yes	yes/yes/yes	yes/yes/yes	–	–	–
3	no/no/no	no/yes/yes	no/yes/yes	yes/yes/yes	yes/yes/yes	yes/yes/yes	–
4	no/no/no	no/yes/yes	no/yes/yes	open/yes/yes	yes/yes/yes	yes/yes/yes	yes/yes/yes
5	no/no/no	no/yes/yes	no/yes/yes	open/yes/yes	open/yes/yes	yes/yes/yes	yes/yes/yes
6	no/no/no	no/yes/yes	no/yes/yes	no/yes/yes	open/yes/yes	yes/yes/yes	yes/yes/yes
7–9	no/no/no	no/yes/yes	no/yes/yes	no/yes/yes	open/yes/yes	open/yes/yes	yes/yes/yes
10	no/no/no	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	open/yes/yes	open/yes/yes
11–14	no/no/no	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	yes/yes/yes	yes/yes/yes
15–17	no/no/no	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	yes/yes/yes
18–19	no/no/no	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	open/yes/yes
20	no/no/no	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	yes/yes/yes
large	no/no/no	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes	no/yes/yes

Table 4: Percolation results and open cases for the BnG/DnG/UnG with integer $k = \beta$, in the format of [BJK+25, Table 1]. For given k, d, "yes" means that the given graph percolates, "no" that it does not, and "open" that the case is at least partially open. Black: [JKLT25]; blue: [CHJK26]; red: [BJK+25]; orange: [LXZ26]; green: this paper (Theorems 2.2, 2.3 and 2.6 with the inputs of Table 2). In the row 7–9 the entry for $k \ge 7$ follows for $d = 7$ also from [BJK+25, Corollary 3.11] with the bounds of [GPS26]. In $d = 10$ and $d \in \{18, 19\}$ the BnG percolates for $k \ge 8$, and $k = 7$ is open in these dimensions. For $d \in \{21, 22, 23\}$ the BnG percolates if and only if $k \ge 8$. The row "large" refers to fixed k and all sufficiently large d; there the BnG does not percolate as soon as $k(k-1) \le 2d$ [JKLT25, Lemma 2.6], and by Theorem 2.5 its critical value of β is $\sqrt{2d} + 1/2 + O(d^{-1/2})$. The red entry for $d = 6$, $k \ge 7$ follows from [BJK+25, Corollary 3.11] with the bounds of [GPS26]. For $d \ge 11$ the green entries use the bounds of [FvdH17, Theorem 1.2] for $d \in \{11,\dots,15,20\}$, together with $p_c(d) \le p_c(15)$ for $d \in \{16,17\}$ and $p_c(d) \le p_c(20)$ for $d \in \{21,22,23\}$.

In high dimensions the integer threshold of the bidirectional graph is determined exactly.

Corollary 2.10. *There is $d_0 \in \mathbb{N}$ such that for all $d \ge d_0$ and $1 \le k \le 2d$ the bidirectional k-neighbour graph on $\mathbb{Z}^d$ percolates if and only if $k(k-1) > 2d$. The same equivalence holds for $d \in \{2,3\} \cup \{11,\dots,17\} \cup \{20,\dots,23\}$.*

The condition has a simple interpretation. Given that the edge $\{x, y\}$ is open, the vertex y keeps each of its other $2d-1$ edges with probability $(k-1)/(2d-1)$, and the corresponding neighbour keeps the edge back with probability $k/(2d)$, so that the expected number of new open edges at y is $k(k-1)/(2d)$. In all cases where percolation of the bidirectional graph is decided, it agrees with this branching heuristic, see Table 4, and so do the simulations for $2 \le d \le 10$.

Conjecture 2.11. *For all $d \geq 2$ and $1 \leq k \leq 2d$, the bidirectional k-neighbour graph on $\mathbb{Z}^d$ percolates if and only if $k(k-1) > 2d$.*

Remark 2.12. Conjecture 2.11 replaces [JKLT25, Conjecture 3.3], which predicts percolation if and only if $k \geq 2\sqrt{d}$. The *only if* part of that conjecture fails in all sufficiently large dimensions, by [BJK+25, Corollary 3.11] and by Corollary 2.10. The first counterexample predicted by Conjecture 2.11 is $(k, d) = (4, 5)$, where $4 < 2\sqrt{5}$ but $k(k-1) = 12 > 10$; it is supported by the numerical value $p_c(5) \approx 0.118$ [MM18], which lies below $\varphi(4)/(10 \cdot 9) = 2/15$. By Theorem 2.2 every open case of Table 4 with $k(k-1) > 2d$ would follow from a rigorous upper bound on $p_c(d)$, see Table 5. Two cases are close: $(k, d) = (5, 5)$ and $(6, 7)$ require improvements of the bounds of [GPS26] by only 3% and 2%.

d	k	$\varphi(k)/(2d(2d-1))$	bound from [GPS26]	numerical $p_c(d)$	required improvement
4	4	0.2143	0.2788	0.1601	23%
5	4	0.1333	0.2284	0.1182	42%
5	5	0.2222	0.2284	0.1182	3%
6	5	0.1515	0.1922	0.0942	21%
7	5	0.1099	0.1682	0.0787	35%
7	6	0.1648	0.1682	0.0787	2%
8	5	0.0833	0.1486	0.0677	44%
8	6	0.1250	0.1486	0.0677	16%
9	5	0.0654	0.1326	0.0595	51%
9	6	0.0980	0.1326	0.0595	26%
10	6	0.0789	0.1206	0.0531	35%
10	7	0.1105	0.1206	0.0531	8%

Table 5: Open cases of the bidirectional k-neighbour graph for $4 \leq d \leq 10$. Each case follows from Theorem 2.2 and a rigorous bound $p_c(d) < \varphi(k)/(2d(2d-1))$. The numerical values of $p_c(d)$ are from [MM18], rounded. The last column is the relative improvement of the bound of [GPS26] needed for this.

3 Local comparison

3.1 Pivotal decomposition for both undirected rules

We treat both undirected graphs at once. Let $\mathrm{R} \in \{\wedge, \vee\}$, and for a local law μ on the subsets of $\mathcal{D}_d$ let $\mathbb{P}_\mu^{\mathrm{R}}$ be the law of the model in which the sets N_x are i.i.d. with law μ and $\{x, x+u\}$ is open if $\mathbb{1}\{u \in N_x\} \,\mathrm{R}\, \mathbb{1}\{-u \in N_{x+u}\}$ holds. Thus $\mathrm{R} = \wedge$ is the bidirectional and $\mathrm{R} = \vee$ the undirected graph. For a random set N with law μ and $A, B \subseteq \mathcal{D}_d$ let

$$\Psi_\mu(A, B) := \mu(N \cap A \neq \emptyset,\ N \cap B \neq \emptyset)\,, \qquad \Psi_\mu(A) := \mu(N \cap A \neq \emptyset)\,.$$

Fix $n \geq 1$ and $a \in B_n$. Let $\mathcal{G}_a$ be the σ-algebra generated by $(N_x)_{x \in B_n \setminus \{a\}}$ and let C_a be the event that o is connected to $\partial B_n \setminus \{a\}$ by an open path in $B_n \setminus \{a\}$, with $C_o := \emptyset$. Let O_a (resp. T_a) be the set of $u \in \mathcal{D}_d$ with $a + u \in B_n$ such that $a + u$ is connected to o (resp. to ∂B_n) by an open path in $B_n \setminus \{a\}$, and let $O_a^* \subseteq O_a$ and $T_a^* \subseteq T_a$ consist of those u for which $a + u$ keeps the edge to a. All these are $\mathcal{G}_a$-measurable.

Lemma 3.1. *Let $a \in B_n \setminus (\{o\} \cup \partial B_n)$. Then $O_a \cap T_a = \emptyset$ on C_a^{c}, and*

$$\mathbb{P}_\mu^{\mathrm{R}}(o \longleftrightarrow \partial B_n \,|\, \mathcal{G}_a) = \mathbb{1}_{C_a} + \mathbb{1}_{C_a^{\mathrm{c}}}\, \mu(E_a)\,,$$

where, for $\mathrm{R} = \wedge$, $E_a = \{N \cap O_a^* \neq \emptyset,\, N \cap T_a^* \neq \emptyset\}$, and, for $\mathrm{R} = \vee$, $E_a = \{O_a^* \neq \emptyset$ or $N \cap O_a \neq \emptyset\} \cap \{T_a^* \neq \emptyset$ or $N \cap T_a \neq \emptyset\}$. For $a = o$ the same holds with the first condition removed, and for $a \in \partial B_n$ with the second condition removed.

Proof. On C_a the event $\{o \longleftrightarrow \partial B_n\}$ occurs whatever N_a is. On C_a^{c}, a direction in $O_a \cap T_a$ would connect o to ∂B_n in $B_n \setminus \{a\}$, hence $O_a \cap T_a = \emptyset$. A self-avoiding open path from o to ∂B_n in B_n must visit a; its vertex before a is $a+v$ with $v \in O_a$ and its vertex after a is $a+u$ with $u \in T_a$, and the edges $\{a, a+v\}$ and $\{a, a+u\}$ are open. Conversely, two such open edges give an open path. The edge $\{a, a+u\}$ is open if $u \in N_a$ and $u \in T_a^* \cup O_a^*$ for $\mathrm{R} = \wedge$, and if $u \in N_a$ or $u \in T_a^* \cup O_a^*$ for $\mathrm{R} = \vee$. This gives the description of E_a, and N_a is independent of $\mathcal{G}_a$. For $a = o$ only an edge towards T_o is needed, and for $a \in \partial B_n$ only an edge towards O_a, since then a itself lies in ∂B_n. □

Figure 4 illustrates the decomposition.

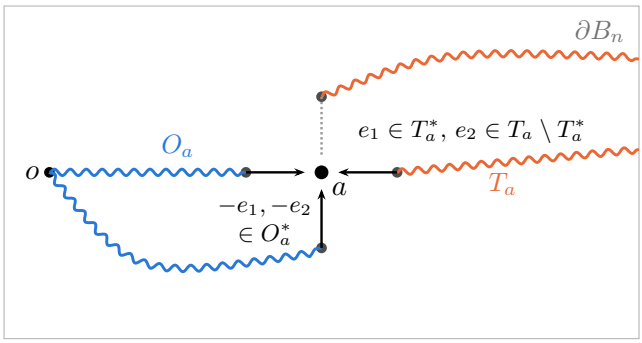


Figure 4: The pivotal decomposition of Lemma 3.1 at a vertex a that is not already bypassed. The neighbours $a - e_1$ and $a - e_2$ are connected to o, and $a + e_1$ and $a + e_2$ to ∂B_n, by open paths avoiding a, so that $O_a = \{-e_1, -e_2\}$ and $T_a = \{e_1, e_2\}$. Arrows indicate neighbours that keep the edge to a; here $O_a^* = \{-e_1, -e_2\}$ and $T_a^* = \{e_1\}$. In the bidirectional graph, o is connected to ∂B_n if and only if N_a meets both O_a^* and T_a^*, an event of type $\Psi(A, B)$ with $|A| = 1$ and $|B| = 2$. In the undirected graph the connection is already present, since O_a^* and T_a^* are nonempty.

In all cases $\mu(E_a)$ equals 0, 1, $\Psi_\mu(A)$ or $\Psi_\mu(A, B)$ for $\mathcal{G}_a$-measurable sets A, B that are disjoint and nonempty, and the case does not depend on μ. As in [BJK+25, Theorem 3.9], replacing the law of one vertex at a time gives the following.

Proposition 3.2. *Let μ, ν be local laws such that, for all disjoint nonempty $A, B \subseteq \mathcal{D}_d$,*

$$\Psi_\mu(A, B) \geq \Psi_\nu(A, B) \qquad \text{and} \qquad \Psi_\mu(A) \geq \Psi_\nu(A)\,. \tag{3.3}$$

Then $\mathbb{P}_\mu^{\mathrm{R}}(o \longleftrightarrow \partial B_n) \geq \mathbb{P}_\nu^{\mathrm{R}}(o \longleftrightarrow \partial B_n)$ for both rules $\mathrm{R} \in \{\wedge, \vee\}$ and all $n \in \mathbb{N}$.

Proof. Enumerate $B_n = \{a_1, \ldots, a_M\}$ and let $\mathbb{P}^{(i)}$ be the law under which $N_{a_1}, \ldots, N_{a_i}$ have law μ and the other sets have law ν. Under $\mathbb{P}^{(i-1)}$ and $\mathbb{P}^{(i)}$ the σ-algebra $\mathcal{G}_{a_i}$ has the same law, and by Lemma 3.1 the difference $\mathbb{P}^{(i)}(o \longleftrightarrow \partial B_n) - \mathbb{P}^{(i-1)}(o \longleftrightarrow \partial B_n)$ is the expectation of $\mathbb{1}_{C_{a_i}^{\mathrm{c}}}$ times $\mu(E_{a_i}) - \nu(E_{a_i})$, which is non-negative by (3.3). □

3.2 The sharp local inequality for mixtures

Let $\pi_{d,q}$ denote the law of the random set that contains every direction independently with probability q. In the model with local law $\pi_{d,q}$ the edges are independent and open with probability q^2 under $\mathrm{R} = \wedge$ and with probability $1-(1-q)^2$ under $\mathrm{R} = \vee$.

Lemma 3.4. *Let $d \in \mathbb{N}$, $\beta \in [0, 2d]$ and $q := \sqrt{\varphi(\beta)/(m(m-1))}$. Then (3.3) holds for $\mu = Q_{d,\beta}$ and $\nu = \pi_{d,q}$, with equality in the first inequality if $|A| = |B| = 1$.*

Proof. Write $\beta = k + \varepsilon$ as above, $u := 1-q$, $\alpha := |A|$ and $\alpha' := |B|$. For $0 \le j \le m$ and $0 \le a \le m$ let $h_j(a) := \binom{m-a}{j}/\binom{m}{j}$ be the probability that a uniform j-subset misses a fixed set of size a, and let $h := (1-\varepsilon)h_k + \varepsilon h_{k+1}$. If $\varphi(\beta) = 0$, that is $\beta \le 1$, then $q = 0$, and if $\beta = m$, then $\mu = \pi_{d,1}$; in both cases there is nothing to prove. Let $1 < \beta < m$, so that $1 \le k \le m-1$ and $\varphi(\beta) > 0$. Since μ and ν are exchangeable, with $G(a) := u^a - h(a)$, inclusion–exclusion gives

$$\Psi_\mu(A) - \Psi_\nu(A) = G(\alpha)\,, \qquad \Psi_\mu(A,B) - \Psi_\nu(A,B) = G(\alpha) + G(\alpha') - G(\alpha+\alpha')\,.$$

Thus we have to show that $G \ge 0$ on $\{0,\dots,m\}$ and that G is subadditive. Let $\Delta(a) := G(a+1) - G(a)$.

Second differences. Since $h_j(a+1) = h_j(a)(m-a-j)/(m-a)$, a direct computation gives $h_j(a+2) - 2h_j(a+1) + h_j(a) = h_j(a)\,j(j-1)/((m-a)(m-a-1))$ for $0 \le a \le m-2$. With $q^2 = \varphi(\beta)/(m(m-1))$ and $\varphi(\beta) = (1-\varepsilon)k(k-1) + \varepsilon(k+1)k$ we obtain

$$\Delta(a+1) - \Delta(a) = q^2\big(u^a - w_1 g_k(a) - w_2 g_{k+1}(a)\big)\,, \qquad g_j(a) := \prod_{i=0}^{a-1} \frac{m-j-i}{m-2-i}\,, \tag{3.5}$$

where $w_1 := (1-\varepsilon)k(k-1)/\varphi(\beta)$ and $w_2 := \varepsilon(k+1)k/\varphi(\beta)$ are non-negative with $w_1 + w_2 = 1$. In particular, $\Delta(1) = \Delta(0)$, which is the equality for $|A| = |B| = 1$.

The case $k = 1$. Then $w_1 = 0$ and $g_2 \equiv 1$, so that $\Delta(a+1) - \Delta(a) = q^2(u^a - 1) \le 0$. Hence G is concave with $G(0) = 0$, and $G(\alpha+\alpha') - G(\alpha') = \sum_{j<\alpha} \Delta(\alpha'+j) \le \sum_{j<\alpha} \Delta(j) = G(\alpha)$. Moreover, $G(1) \ge 0$ by (3.6) below and $G(m) = u^m \ge 0$, so that $G \ge 0$ by concavity.

The case $k \ge 2$. Since $g_{k+1}(a) = g_k(a)\,(m-k-a)/(m-k)$, the bracket in (3.5) equals $u^a\big(1 - r(a)\big)$ with

$$r(a) := u^{-a} g_k(a)\big(1 - ca\big)\,, \qquad c := \frac{w_2}{m-k} \in \Big[0, \frac{1}{m-k}\Big]\,.$$

We have $r(0) = 1$, and while $r(a) > 0$ (which forces $a \le m-k$ and $ca < 1$),

$$\frac{r(a+1)}{r(a)} = \frac{1}{u}\Big(1 - \frac{k-2}{m-2-a}\Big)\Big(1 - \frac{c}{1-ca}\Big)\,.$$

For $a < m-k$ both factors in brackets are positive and non-increasing in a; for $a = m-k$ the first factor vanishes, so that $r(a+1) = 0$, and r stays 0 once it vanishes. Hence r is first non-decreasing and then non-increasing, and $\{a\colon r(a) \ge 1\} = \{0,\dots,a_*\}$ for some $0 \le a_* \le m-2$. By (3.5), Δ is non-increasing on $\{0,\dots,a_*+1\}$ and increasing on $\{a_*+1,\dots,m-1\}$. Since $h_j(m-1) = h_j(m) = 0$ for $j \ge 2$, we have $\Delta(m-1) = -qu^{m-1} < 0$, and therefore $\Delta < 0$ on $\{a_*+1,\dots,m-1\}$. Let $b := \min\{a\colon \Delta(a) \le 0\} \le a_*+1$. Then $\Delta(0),\dots,\Delta(b-1)$ are positive and non-increasing, and $\Delta(a) \le 0$ for $a \ge b$. Consequently G is non-decreasing with non-increasing increments on $\{0,\dots,b\}$ and non-increasing on $\{b,\dots,m\}$. Since $G(0) = 0$ and

$G(m) = u^m \geq 0$, this gives $G \geq 0$. Subadditivity follows: if $\alpha + \alpha' \leq b$ use the non-increasing increments; if $\max\{\alpha, \alpha'\} \geq b$ use $G(\alpha + \alpha') \leq G(\max\{\alpha, \alpha'\})$ and $G \geq 0$; otherwise $G(\alpha + \alpha') \leq G(b) \leq G(\alpha) + G(b - \alpha) \leq G(\alpha) + G(\alpha')$, since $1 \leq b - \alpha < \alpha' < b$.

The value $G(1)$. We have $G(1) = u - h(1) = \beta/m - q$, and

$$(m-1)\beta^2 - m\,\varphi(\beta) = k(m-k-2\varepsilon) + (m-1)\varepsilon^2 \geq 0\,, \tag{3.6}$$

because $m - k - 2\varepsilon > 0$ for $k \leq m - 2$ and the left-hand side equals $(m-1)(1-\varepsilon)^2$ for $k = m - 1$. Hence $q^2 = \varphi(\beta)/(m(m-1)) \leq (\beta/m)^2$ and $G(1) \geq 0$. □

Remark 3.7. Lemma 3.4 has also been checked numerically in 80-digit arithmetic for all $m \leq 40$, all k, and seven values of ε, about $2.3 \cdot 10^6$ pair inequalities, see `code/check_mixture_inequality.py`.

Proof of Theorem 2.2. Combine Proposition 3.2 with Lemma 3.4 for both rules. The statement for the directed graph follows from the local inequality [BJK+25, Corollary 3.3], the local-to-global comparison [BJK+25, Theorem 2.1], and the identity between directed and undirected Bernoulli one-arm probabilities [BJK+25, Lemma 3.5], see [BJK+25, Corollary 3.7]. □

4 Absence of percolation

We prove Theorem 2.3 by bounding the expected number of open self-avoiding paths from the origin. For the bidirectional and the directed graph the vertices of a path contribute independent factors; for the undirected graph consecutive edges share a vertex, and a 2×2 transfer matrix accounts for this dependence.

Proof of Theorem 2.3. Let $c_n(d)$ be the number of self-avoiding paths of length n from o, so that $c_n(d) \leq (c(d) + \delta)^n$ for every $\delta > 0$ and n large. A self-avoiding path $(x_0, \dots, x_n)$ involves the sets $N_{x_0}, \dots, N_{x_n}$ of distinct vertices, which are independent.

(B) The path is open in the bidirectional graph if x_0 and x_n keep their path edge and each x_i, $0 < i < n$, keeps both of its path edges. This has probability $(\beta/m)^2\big(\varphi(\beta)/(m(m-1))\big)^{n-1}$. If $c(d)\varphi(\beta) < m(m-1)$, the expected number of open self-avoiding paths of length n from o tends to zero, and $\theta_d^{\mathrm{B}}(\beta) = 0$. Since $c(d) < 2d - 1$ for $d \geq 2$ [JKLT25, Section 2.3], the condition holds if $\varphi(\beta) \leq m$.

(D) A directed path is open with probability $(\beta/m)^n$, and the same argument applies, see [CHJK26, Lemma 2.2].

(U) For $\beta \leq 1$ every vertex keeps at most one edge, and the undirected graph is a subgraph of the undirected one-neighbour graph in the coupling of Section 2, which does not percolate by [JKLT25, Proposition 2.5]. For the second condition, let $f_i \in \{0, 1\}$ indicate that x_i keeps the edge $\{x_i, x_{i+1}\}$. The edge $\{x_{i-1}, x_i\}$ is open if $f_{i-1} = 1$ or x_i keeps $\{x_i, x_{i-1}\}$. For $0 < i < n$ the conditional probability, given f_{i-1}, that x_i satisfies this requirement and has $f_i = f$ is the entry (f_{i-1}, f) of

$$M := \begin{pmatrix} r - \kappa & \kappa \\ 1 - r & r \end{pmatrix}, \qquad r := \beta/m\ :$$

if $f_{i-1} = 1$ there is no requirement, and if $f_{i-1} = 0$ the vertex x_i has to keep the backward edge, which together with the forward edge has probability κ and without it $r - \kappa$. Hence the probability that the path is open is at most $\mathbf{1}^\top M^{n-1}\mathbf{1} \leq C\lambda^n$ for some $C < \infty$, where

$$\lambda = r - \frac{\kappa}{2} + \sqrt{\kappa\Big(\frac{\kappa}{4} + 1 - r\Big)} \leq \bar{\lambda}_d(\beta)$$

is the Perron root of M; the inequality uses $\kappa \le r^2 \le 4r$, see (3.6). The claim follows as before. □

5 Proofs of the main results

Proof of Theorem 2.4. The lower bounds are Theorem 2.3, and the upper bounds follow from Theorem 2.2, since the comparison models percolate above $p_c(d)$ and φ is increasing on $[1, 2d]$. For Table 2 we use $p_c(2) = 1/2$ [Kes80]; for $3 \le d \le 10$ the bounds $p_c(d) \le p^*(d)$ of [GPS26, Theorem 2.1]; the bounds of [FvdH17, Theorem 1.2] for $d \in \{11, \dots, 15, 20\}$, together with $p_c(d) \le p_c(d')$ for $d \ge d'$; the bounds $c(2) \le 2.679192495$, $c(3) \le 4.7387$, $c(4) \le 6.8040$, $c(5) \le 8.8602$ of [PT00] and $c(d) < 2d - 1$ otherwise; and for the directed graph also $\beta_c^{\mathrm{D}}(d) \le 2$ [CHJK26, LXZ26] and the bounds of Theorem 2.7 for $3 \le d \le 10$. For the undirected lower bound we use that $\bar\lambda_d$ is increasing on $[1, 2d]$, since its derivative is $1/m + \frac{1}{2}\,\partial_\beta\kappa\,(\kappa^{-1/2} - 1) > 0$, and solve $c(d)\bar\lambda_d(\beta) = 1$ by bisection. See `code/rigorous_bounds.py`. □

Proof of Corollary 2.10. Let $d \ge 2$. If $k(k-1) \le 2d$, then $\varphi(k) = k(k-1) \le 2d$, and Theorem 2.3(B) gives absence of percolation. Conversely, since $k(k-1)$ and $2d$ are even, $k(k-1) > 2d$ implies $\varphi(k) \ge 2d+2$. As in the proof of Theorem 2.5 below, $2d(2d-1)p_c(d) \le 2d + C/(2d) < 2d+2$ for all large d, so that Theorem 2.2 gives percolation. The explicit dimensions follow from Theorem 2.6 for $d = 2$ and from the bounds of Table 2 and [FvdH17, Theorem 1.2], see Table 4; for $d = 16, 17$ and $d = 21, 22, 23$ we use $p_c(d) \le p_c(15)$ and $p_c(d) \le p_c(20)$, respectively. □

Proof of Theorem 2.5. Bidirectional graph. By Theorem 2.3(B), $\beta_c^{\mathrm{B}}(d) \ge \varphi^{-1}(2d)$. Let k be the largest integer with $k(k-1) \le 2d$. Then $\varphi^{-1}(2d) = k + (2d - k(k-1))/(2k) = (k+1)/2 + d/k$, and with $x_0 := \sqrt{2d}$,

$$\varphi^{-1}(2d) - \Big(x_0 + \frac{1}{2}\Big) = \frac{(k - x_0)^2}{2k} \in \Big[0, \frac{1}{2k}\Big],$$

since $|k - x_0| \le 1$. This proves the first chain of inequalities and the second equality for β_c^{B}, once the first is shown. By [HS95] and [vdHS06, Theorem 1.1], $p_c(d) = (2d)^{-1} + (2d)^{-2} + O((2d)^{-3})$, so that $m(m-1)p_c(d) \le m + C/m$ for some $C < \infty$ and all large d. On $[k, \infty)$ the function φ has slope at least $2k \ge \sqrt{m}$, hence

$$\beta_c^{\mathrm{B}}(d) \le \varphi^{-1}(m + C/m) \le \varphi^{-1}(m) + \frac{C}{m^{3/2}}.$$

Directed graph. By Theorem 2.3(D) and $c(d) < 2d - 1$, $\beta_c^{\mathrm{D}}(d) \ge 2d/(2d-1) = 1 + 1/(2d-1)$, and by Theorem 2.2, $\beta_c^{\mathrm{D}}(d) \le 2d\,p_c(d) = 1 + 1/(2d) + O(d^{-2})$ by [HS95].

Undirected graph. The upper bound follows from (2.1). For the lower bound let $\beta = 1 + \varepsilon$ with $\varepsilon \in (0, 1)$, so that $\varphi(\beta) = 2\varepsilon$ and $\kappa = 2\varepsilon/(m(m-1))$. Since $c(d) < m - 1$,

$$c(d)\,\bar\lambda_d(\beta) < (m-1)\Big(\frac{1+\varepsilon}{m} + \sqrt{\frac{2\varepsilon}{m(m-1)}}\Big) \le 1 - \frac{1}{m} + \varepsilon + \sqrt{2\varepsilon},$$

which is at most 1 if $\varepsilon + \sqrt{2\varepsilon} \le 1/m$, that is, if $\sqrt{\varepsilon} \le (\sqrt{2 + 4/m} - \sqrt{2})/2$. Hence Theorem 2.3(U) gives the lower bound, and $((\sqrt{2 + 2/d} - \sqrt{2})/2)^2 = (1 + O(d^{-1}))/(8d^2)$. □

6 The directed graph: column projection and comparison of dimensions

In this section we prove Theorems 2.7 and 2.9. The method reduces a directed model on $\mathbb{Z}^d$ to a directed model on $\mathbb{Z}^2$ by letting the vertices of a column act together as one vertex of the lattice one dimension lower. It is inspired by the coupling in the proof of [JKLT25, Theorem 2.3], and the same reduction was found independently in [LXZ26], where it is used for $\beta = 2$. For a local law μ on the subsets of $\mathcal{D}_d$ we write $\mathbb{P}^{\mathrm{D}}_\mu$ for the directed model, in which every vertex x independently draws N_x with law μ and sends a directed edge to every $x+u$ with $u \in N_x$, and $\theta^{\mathrm{D}}_d(\mu)$ for the probability that the out-cluster $\mathcal{C}(o)$ of the origin is infinite. In particular, $\theta^{\mathrm{D}}_d(\beta) = \theta^{\mathrm{D}}_d(Q_{d,\beta})$ in the notation of Section 2.

6.1 The planar criterion

For $p \in [0,1]$ the law $\pi_{d,p}$ of Section 3 defines the directed i.i.d. model, which has the same one-arm probabilities as Bernoulli bond percolation with parameter p [BJK$^+$25, Lemma 3.5]. If two local laws satisfy $\mu(N \cap A \neq \emptyset) \le \mu'(N \cap A \neq \emptyset)$ for all nonempty $A \subseteq \mathcal{D}_d$, then the one-arm probabilities of the directed models satisfy $\mathbb{P}^{\mathrm{D}}_\mu(o \longleftrightarrow \partial B_n) \le \mathbb{P}^{\mathrm{D}}_{\mu'}(o \longleftrightarrow \partial B_n)$ for all n [BJK$^+$25, Theorem 2.1]. A law ν on the subsets of $\mathcal{D}_d$ is *exchangeable* if it is invariant under all permutations of $\mathcal{D}_d$. Then $t_\ell := \nu(N \cap A = \emptyset)$ depends only on $\ell = \#A$, and we call $t = (t_0, \dots, t_{2d})$ the *avoidance vector* of ν; it determines ν.

Lemma 6.1. *Let ν be an exchangeable law on the subsets of $\mathcal{D}_2$ with avoidance vector t. If $t_\ell \le (1-p)^\ell$ for $\ell = 1, \dots, 4$ and some $p \in (1/2, 1]$, then $\theta^{\mathrm{D}}_2(\nu) \ge \theta^{\mathrm{bond}}_2(p) > 0$.*

Proof. Since $\pi_{2,p}(N \cap A = \emptyset) = (1-p)^{\#A}$, the assumption is the comparison above for $\mu = \pi_{2,p}$ and $\mu' = \nu$. Together with the identification with bond percolation and [Kes80] this gives the claim. □

6.2 The column projection

Let $d \ge 3$ and write $x = (y, z) \in \mathbb{Z}^{d-1} \times \mathbb{Z}$. We identify $\mathcal{D}_{d-1}$ with the horizontal directions in $\mathcal{D}_d$; the vertical directions are $\pm e_d$. Fix a configuration $(N_x)_{x \in \mathbb{Z}^d}$.

Definition 6.2. For $y \in \mathbb{Z}^{d-1}$ and $h \in \mathbb{Z}$, the *explored set* $E(y,h) \subseteq \mathbb{Z}$ is the smallest set containing h such that, for every $i \ge 0$, the conditions $h+i \in E(y,h)$ and $e_d \in N_{(y,h+i)}$ imply $h+i+1 \in E(y,h)$, and the conditions $h-i \in E(y,h)$ and $-e_d \in N_{(y,h-i)}$ imply $h-i-1 \in E(y,h)$. The *collected set* is

$$M(y,h) := \bigcup_{z \in E(y,h)} \left(N_{(y,z)} \cap \mathcal{D}_{d-1}\right).$$

For $u \in M(y,h)$, the *entry height* $\eta(y,h,u)$ is the element $z \in E(y,h)$ with $u \in N_{(y,z)}$ that minimises $|z-h|$, with ties broken towards larger z. The law of $M(y,h)$ is denoted by $\Phi_d(\mu)$.

The set $E(y,h)$ is an interval containing h. Above h it grows as long as the current top vertex keeps $+e_d$, below h as long as the current bottom vertex keeps $-e_d$; a direction pointing back towards h plays no role. Since $M(y,h)$ is a fixed function of the shifted column $(N_{(y,h+z)})_{z \in \mathbb{Z}}$, its law does not depend on (y,h). Figure 5 shows one step.

Proposition 6.3. *For every $d \ge 3$ and every local law μ on the subsets of $\mathcal{D}_d$ we have $\theta^{\mathrm{D}}_d(\mu) \ge \theta^{\mathrm{D}}_{d-1}(\Phi_d(\mu))$.*

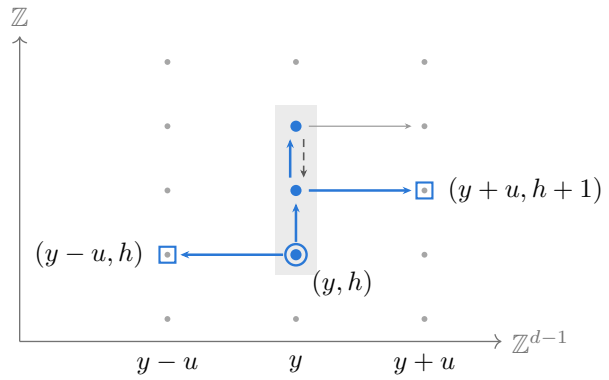


Figure 5: Schematic column exploration from (y,h); the horizontal axis represents $\mathbb{Z}^{d-1}$ by a single direction u. The vertex (y,h) keeps $+e_d$ and $-u$, the vertex $(y,h+1)$ keeps $+e_d$ and $+u$, and the vertex $(y,h+2)$ keeps $-e_d$, which plays no role (dashed), and $+u$. The explored set (shaded) is $E(y,h)=\{h,h+1,h+2\}$, the collected set is $M(y,h)=\{-u,+u\}$, and the squares mark the entry points $(y-u,h)$ and $(y+u,h+1)$ from which the neighbouring columns are explored.

Proof. Fix an enumeration of $\mathbb{Z}^{d-1}$. We run the following exploration, which maintains a set D of discovered vertices of $\mathbb{Z}^{d-1}$ with heights $h(y)\in\mathbb{Z}$, $y\in D$, and a set $S\subseteq D$ of processed vertices. Initially $D=\{o\}$, $h(o)=0$ and $S=\emptyset$. In step $i\geq 1$, if $D\setminus S$ is empty, the exploration stops. Otherwise let y_i be the element of $D\setminus S$ with the smallest index, set $M_i:=M(y_i,h(y_i))$, add y_i to S, and add every y_i+u with $u\in M_i$ and $y_i+u\notin D$ to D, with height $h(y_i+u):=\eta(y_i,h(y_i),u)$.

Lifting. We show that $(y,h(y))\in\mathcal{C}(o)$ for every $y\in D$, by induction over the order of discovery. This holds for $y=o$. If y_i+u is discovered in step i and $z:=\eta(y_i,h(y_i),u)$, then $(y_i,h(y_i))$ reaches (y_i,z) along the explored interval, and (y_i,z) has an edge to $(y_i+u,z)=(y_i+u,h(y_i+u))$. Since $y\mapsto(y,h(y))$ is injective, $\#\mathcal{C}(o)\geq\#D$.

Law of D. Let $\mathcal{F}_i$ be the σ-algebra generated by $y_1,\dots,y_i$ and the sets $N_{(y_l,z)}$, $l\leq i$, $z\in\mathbb{Z}$. Whether the exploration stops after step i, and otherwise y_{i+1} and $h(y_{i+1})$, are $\mathcal{F}_i$-measurable. For fixed distinct $z_1,\dots,z_i,y\in\mathbb{Z}^{d-1}$, the event $\{y_1=z_1,\dots,y_i=z_i,\,y_{i+1}=y\}$ depends only on the columns above $z_1,\dots,z_i$, which are independent of the column $\{y\}\times\mathbb{Z}$. Hence, conditionally on $\mathcal{F}_i$ and on $y_{i+1}=y$, the sets in $\{y\}\times\mathbb{Z}$ are i.i.d. with law μ, and M_{i+1} has law $\Phi_d(\mu)$. By induction over i, the sequence $(y_i,M_i)_i$ has the same law as the sequence produced by the same rule from an i.i.d. family $(\xi_y)_{y\in\mathbb{Z}^{d-1}}$ with law $\Phi_d(\mu)$, where ξ_{y_i} replaces M_i; the heights do not enter the evolution of D and S. In the latter exploration every discovered vertex is processed after finitely many steps, and processing y discovers all its out-neighbours $y+\xi_y$. Hence D has the law of the out-cluster of the origin in the directed model on $\mathbb{Z}^{d-1}$ with local law $\Phi_d(\mu)$, and $\mathbb{P}(\#D=\infty)=\theta^{\mathrm{D}}_{d-1}(\Phi_d(\mu))$. Together with the lifting this proves the claim. □

Applying Proposition 6.3 successively in dimensions $d,d-1,\dots,3$ gives

$$\theta^{\mathrm{D}}_d(\mu)\geq\theta^{\mathrm{D}}_2\big(\Phi_3\circ\cdots\circ\Phi_d(\mu)\big)\,. \tag{6.4}$$

6.3 The avoidance recursion

For $j \in \mathbb{N}$ let $\mathcal{V}_j := \{t \in [0,1]^{j+1} \colon 1 = t_0 > t_1 \geq t_2 \geq \cdots \geq t_j\}$. Avoidance vectors of exchangeable laws with $\nu(N = \emptyset) < 1$ lie in $\mathcal{V}_{2d}$. For $j \geq 4$ and $t \in \mathcal{V}_j$ let $\Gamma(t)_0 := 1$ and, for $1 \leq \ell \leq j-2$,

$$\psi_\ell := \frac{t_{\ell+1}}{1 - t_\ell + t_{\ell+1}}\,, \qquad \Gamma(t)_\ell := (1-\psi_\ell)^2\, t_{\ell+2} + 2\psi_\ell(1-\psi_\ell)\, t_{\ell+1} + \psi_\ell^2\, t_\ell\,. \tag{6.5}$$

The denominator is at least $1 - t_\ell \geq 1 - t_1 > 0$.

Lemma 6.6. *Let $j \geq 4$. The map Γ sends $\mathcal{V}_j$ to $\mathcal{V}_{j-2}$, and $t_{\ell+2} \leq \Gamma(t)_\ell \leq t_\ell$ for $1 \leq \ell \leq j-2$. Moreover, Γ is non-decreasing: if $t, t' \in \mathcal{V}_j$ with $t \leq t'$ componentwise, then $\Gamma(t) \leq \Gamma(t')$.*

Proof. On the set $\mathcal{T} := \{(r_0, r_1, r_2) \colon 1 > r_0 \geq r_1 \geq r_2 \geq 0\}$ let $g(r_0, r_1) := r_1/(1 - r_0 + r_1)$ and $H(r_0, r_1, r_2) := (1-g)^2 r_2 + 2g(1-g)r_1 + g^2 r_0$ with $g = g(r_0, r_1)$, so that $\Gamma(t)_\ell = H(t_\ell, t_{\ell+1}, t_{\ell+2})$ for $\ell \geq 1$; these triples lie in $\mathcal{T}$ because $t_\ell \leq t_1 < 1$. The three weights in H are non-negative and sum to one, which gives $r_2 \leq H \leq r_0$. The function g is non-decreasing in r_0 and in r_1, and

$$\frac{\partial H}{\partial g} = 2(1-g)(r_1 - r_2) + 2g(r_0 - r_1) \geq 0\,.$$

Together with the non-negative weights, all partial derivatives of H are non-negative on $\mathcal{T}$. If $(r_0, r_1, r_2) \leq (r_0', r_1', r_2')$ both lie in $\mathcal{T}$, raising first r_0, then r_1 and then r_2 keeps the triple in $\mathcal{T}$, so $H(r_0, r_1, r_2) \leq H(r_0', r_1', r_2')$. This proves that Γ is non-decreasing. Applied to $(t_{\ell+1}, t_{\ell+2}, t_{\ell+3}) \leq (t_\ell, t_{\ell+1}, t_{\ell+2})$ it gives $\Gamma(t)_{\ell+1} \leq \Gamma(t)_\ell$, and $\Gamma(t)_1 \leq t_1 < 1$ shows $\Gamma(t) \in \mathcal{V}_{j-2}$. □

Lemma 6.7. *Let $d \geq 3$ and let μ be an exchangeable law on the subsets of $\mathcal{D}_d$ with avoidance vector $t \in \mathcal{V}_{2d}$. Then $\Phi_d(\mu)$ is exchangeable with avoidance vector $\Gamma(t) \in \mathcal{V}_{2d-2}$.*

Proof. The exploration in Definition 6.2 treats all horizontal directions alike, so $\Phi_d(\mu)$ is exchangeable. Fix $A \subseteq \mathcal{D}_{d-1}$ with $\#A = \ell \in \{1, \ldots, 2d-2\}$ and consider the exploration from $(o, 0)$. Let ϱ be the probability that no vertex (o, z) with $z \geq 1$ in $E(o, 0)$ keeps a direction in A, given that $1 \in E(o, 0)$; it does not depend on the root. The vertex $(o, 1)$ avoids $A \cup \{e_d\}$ with probability $t_{\ell+1}$, and then the exploration stops above; it avoids A and keeps e_d with probability $t_\ell - t_{\ell+1}$, and then the exploration continues from $(o, 2)$ with fresh sets; its choice of $-e_d$ plays no role. Hence $\varrho = t_{\ell+1} + (t_\ell - t_{\ell+1})\varrho$, and since $t_\ell - t_{\ell+1} \leq t_1 < 1$, the unique solution is $\varrho = \psi_\ell$. By symmetry the same holds below the root. At the root, exchangeability and inclusion–exclusion show that the probabilities to avoid A and to keep neither, exactly one, or both of $\pm e_d$ are $t_{\ell+2}$, $2(t_{\ell+1} - t_{\ell+2})$ and $t_\ell - 2t_{\ell+1} + t_{\ell+2}$. Given the root, the parts above and below are independent, and therefore

$$\Phi_d(\mu)(N \cap A = \emptyset) = t_{\ell+2} + 2(t_{\ell+1} - t_{\ell+2})\,\psi_\ell + (t_\ell - 2t_{\ell+1} + t_{\ell+2})\,\psi_\ell^2 = \Gamma(t)_\ell\,.$$

For $\ell = 0$ both sides equal one, and $\Gamma(t) \in \mathcal{V}_{2d-2}$ by Lemma 6.6. □

For $\beta \in [1, 2]$ the law $Q_{d,\beta}$ is exchangeable with avoidance vector

$$\tau_\ell^{(d)}(\beta) = (2-\beta)\,\frac{2d-\ell}{2d} + (\beta - 1)\,\frac{(2d-\ell)(2d-\ell-1)}{2d(2d-1)}\,, \qquad 0 \leq \ell \leq 2d\,, \tag{6.8}$$

which is non-increasing in β for every ℓ. Since Γ is non-decreasing, rounding all entries upwards after each application of Γ yields rigorous upper bounds on iterated avoidance vectors; all computer-assisted statements below use exact rational arithmetic with this rounding.

6.4 Comparison with proper subsets

The local-to-global comparison of [BJK+25, Theorem 2.1] requires a local inequality for all nonempty sets of directions. Its proof only uses proper subsets, except at the origin, and the origin can be handled separately.

Proposition 6.9. *Let μ, ν be local laws on the subsets of $\mathcal{D}_d$ with*

$$\mu(N \cap A \neq \emptyset) \leq \nu(N \cap A \neq \emptyset) \qquad \text{for all } \emptyset \neq A \subsetneq \mathcal{D}_d\,. \tag{6.10}$$

If $\theta_d^{\mathrm{D}}(\mu) > 0$, then $\theta_d^{\mathrm{D}}(\nu) > 0$.

Proof. Let ν^o be the law of the model in which the origin uses μ and every other vertex uses ν. Fix n, and for $U \subseteq B_n \setminus \{o\}$ let $\mathbb{P}^U$ be the law in which the vertices in U use ν and all others use μ. We show $\mathbb{P}^U(o \longleftrightarrow \partial B_n) \leq \mathbb{P}^{U\cup\{a\}}(o \longleftrightarrow \partial B_n)$ for $a \in B_n \setminus (U \cup \{o\})$, where $\longleftrightarrow$ refers to directed paths in B_n. Condition on all sets except N_a. Either the event $\{o \longleftrightarrow \partial B_n\}$ does not depend on N_a, or $o \not\longleftrightarrow \partial B_n$ when $N_a = \emptyset$ while $o \longleftrightarrow a$. In the second case let A be the set of directions u such that $a + u \longleftrightarrow \partial B_n$ without using a; then the event equals $\{N_a \cap A \neq \emptyset\}$, since a path from o to ∂B_n through a leaves a for the last time towards a vertex in $a + A$. Moreover $A \neq \mathcal{D}_d$: the last vertex w before a on a path from o to a that visits a only at its end satisfies $o \longleftrightarrow w$ without using a, so $w - a \in A$ would give $o \longleftrightarrow \partial B_n$ with $N_a = \emptyset$. If $A = \emptyset$, the event fails for every N_a. Otherwise (6.10) gives the claim. Adding the vertices of $B_n \setminus \{o\}$ one by one and letting $n \to \infty$ yields $\theta_d^{\mathrm{D}}(\nu^o) \geq \theta_d^{\mathrm{D}}(\mu) > 0$.

On the event that the out-cluster of o is infinite under ν^o, an infinite self-avoiding path starts with an edge to some $o + u$ and continues without returning to o. Writing G_u for the event that $o + u$ has an infinite out-path avoiding o, which depends only on vertices other than o and has the same probability under ν^o and ν, we obtain $0 < \theta_d^{\mathrm{D}}(\nu^o) \leq \sum_u \mu(u \in N)\, \mathbb{P}_\nu^{\mathrm{D}}(G_u)$. Choose u with $\mu(u \in N)\mathbb{P}_\nu^{\mathrm{D}}(G_u) > 0$. By (6.10) with $A = \{u\}$, $\nu(u \in N) \geq \mu(u \in N) > 0$, and independence gives $\theta_d^{\mathrm{D}}(\nu) \geq \nu(u \in N)\, \mathbb{P}_\nu^{\mathrm{D}}(G_u) > 0$. □

Proposition 6.11 (Comparison principle)**.** *Let $d \geq 2$, $s \geq 1$ and $\beta \in [1, 2]$ be such that*

$$\big(\Gamma^{\circ s}(\tau^{(d+s)}(\beta))\big)_\ell \leq \tau_\ell^{(d)}(\beta) \qquad \text{for all } 1 \leq \ell \leq 2d - 1\,. \tag{6.12}$$

If $\theta_d^{\mathrm{D}}(\beta) > 0$, then $\theta_{d+s}^{\mathrm{D}}(\beta) > 0$.

Proof. By Lemma 6.7, $\nu := \Phi_{d+1} \circ \cdots \circ \Phi_{d+s}(Q_{d+s,\beta})$ is exchangeable with avoidance vector $\Gamma^{\circ s}(\tau^{(d+s)}(\beta))$, and (6.12) is exactly (6.10) for $\mu = Q_{d,\beta}$ and this ν. Hence $\theta_d^{\mathrm{D}}(\nu) > 0$ by Proposition 6.9, and Proposition 6.3, applied s times, gives $\theta_{d+s}^{\mathrm{D}}(\beta) \geq \theta_d^{\mathrm{D}}(\nu) > 0$. □

Condition (6.12) only involves sets of at most $2d - 1$ directions; this is why Proposition 6.9 is needed.

6.5 Consecutive dimensions

Lemma 6.13. *For all $d \geq 3$, $\beta \in [1.45, 2]$ and $0 \leq \ell \leq 2d - 2$ we have $\Gamma(\tau^{(d+1)}(\beta))_\ell \leq \tau_\ell^{(d)}(\beta)$.*

Proof. Write $\varepsilon := \beta - 1$, $r := 2d + 2 \geq 8$ and $a := r - \ell \in \{4, \ldots, r\}$. For $\ell = 0$ both sides equal one, so let $4 \leq a \leq r - 1$. Inserting (6.8) into (6.5) and simplifying gives

$$\tau_\ell^{(d)}(\beta) - \Gamma(\tau^{(d+1)}(\beta))_\ell = \frac{2(r-a)\,P(a,r,\varepsilon)}{r(r-1)(r-2)(r-3)\big((r-1)^2 + \varepsilon(r+1-2a)\big)^2}\,, \tag{6.14}$$

where P is a polynomial of degree three in a and in ε and of degree five in r, which the script `certify_consecutive_dimensions.py` prints and checks against (6.14). The denominator is positive since $(r-1)^2+\varepsilon(r+1-2a) \geq (r-1)^2-(r-3) > 0$. It remains to show $P \geq 0$. Substitute $\varepsilon = 9/20 + (11/20)\sigma$ with $\sigma \in [0,1]$ and expand P in the Bernstein basis of degree three in σ. On the region $a \geq 7$, written as $a = 7+x$, $r = 8+x+y$ with $x, y \geq 0$, and on the three lines $a \in \{4,5,6\}$, $r = 8+y$ with $y \geq 0$, every Bernstein coefficient is a polynomial in x, y with non-negative coefficients, as verified by computer algebra. These regions cover all pairs $4 \leq a \leq r-1$ with $r \geq 8$, so $P \geq 0$. □

For $\beta = 2$ the polynomial P factorises, and Lemma 6.13 then holds for all $d \geq 2$ with a short proof by hand, see also [LXZ26, Lemma 2.4]. For $\beta = 1.3$ the inequality fails in every dimension $3 \leq d \leq 300$ that we tested, mostly at $\ell = 2d-2$.

The two largest set sizes cannot be compared in this way, because the projected out-set has size at most one with positive probability. We absorb them by a closure: for $t \in \mathcal{V}_j$ let $(\mathcal{K}t)_\ell := t_\ell$ for $\ell \leq j-2$ and $(\mathcal{K}t)_{j-1} = (\mathcal{K}t)_j := t_{j-2}$, and set

$$\omega^{(d)}(\beta) := (\mathcal{K} \circ \Gamma)^{\circ(d-2)}\big(\mathcal{K}\tau^{(d)}(\beta)\big) \in \mathcal{V}_4 .$$

Lemma 6.15. *Let $\beta \in [1.45, 2]$. Then $\Gamma^{\circ(d-2)}(\tau^{(d)}(\beta)) \leq \omega^{(d)}(\beta)$ for $d \geq 3$, and $d \mapsto \omega^{(d)}(\beta)$ is non-increasing on $\{3, 4, \dots\}$.*

Proof. The first claim follows from $t \leq \mathcal{K}t$ and the monotonicity of Γ and $\mathcal{K}$. For the second, $\Gamma(\mathcal{K}\tau^{(d+1)})_\ell = \Gamma(\tau^{(d+1)})_\ell \leq \tau^{(d)}_\ell$ for $\ell \leq 2d-2$ by Lemma 6.13, since $\mathcal{K}$ only changes the two largest indices and $\Gamma(t)_\ell$ only depends on $t_\ell, t_{\ell+1}, t_{\ell+2}$. As $\mathcal{K}$ on $\mathcal{V}_{2d}$ only reads indices up to $2d-2$, this gives $(\mathcal{K} \circ \Gamma)(\mathcal{K}\tau^{(d+1)}) \leq \mathcal{K}\tau^{(d)}$, and the claim follows by applying the non-decreasing map $(\mathcal{K} \circ \Gamma)^{\circ(d-2)}$. □

6.6 Proofs of Theorems 2.7 and 2.9

Proof of Theorem 2.7. For each pair $(d, \bar\beta_d)$ in Table 3 with $d \leq 30$ we apply Γ exactly $d-2$ times to $\tau^{(d)}(\bar\beta_d)$ in rational arithmetic and round every entry upwards to the grid $2^{-100}\mathbb{Z}$ after each step, which keeps the vectors ordered. By Lemmas 6.6 and 6.7, the result bounds the avoidance vector of $\Phi_3 \circ \cdots \circ \Phi_d(Q_{d,\bar\beta_d})$ from above. In every case its entries satisfy $t_\ell < 2^{-\ell}$ for $\ell = 1, \dots, 4$, see `code/verify_uniform_bounds_and_plane.py`, so that Lemma 6.1 applies with some $p > 1/2$, and (6.4) gives $\theta^{\mathrm{D}}_d(\bar\beta_d) > 0$. By monotonicity in β, $\beta^{\mathrm{D}}_c(d) \leq \bar\beta_d$.

For $d \geq 31$ let $\beta = 1.457$. The upward-rounded vector $\omega^{(31)}(\beta)$ satisfies $t_\ell < 2^{-\ell}$ for $\ell = 1, \dots, 4$, and by Lemma 6.15, $\Gamma^{\circ(d-2)}(\tau^{(d)}(\beta)) \leq \omega^{(d)}(\beta) \leq \omega^{(31)}(\beta)$ for every $d \geq 31$. As before, $\beta^{\mathrm{D}}_c(d) \leq 1.457$. The suprema follow from the table, since $\bar\beta_d \leq 1.477$ for $d \geq 8$ and $\bar\beta_d \leq 1.447$ for $9 \leq d \leq 30$. □

Proof of Theorem 2.9. For $d \geq 8$, Theorem 2.7 and (2.8) give $\beta^{\mathrm{D}}_c(d) \leq 1.477 < 1.4929 \leq \beta^{\mathrm{D}}_c(2)$. For $d = 7$ let $\beta > \beta^{\mathrm{D}}_c(2)$. If $\beta \geq \bar\beta_7 = 1.512$, then $\theta^{\mathrm{D}}_7(\beta) > 0$ by Theorem 2.7. Otherwise $\beta \in [1.4929, 1.512]$ by (2.8). Split this interval into 40 subintervals $[\beta_1, \beta_2]$. Since $\tau^{(d)}(\beta)$ is non-increasing in β and Γ is monotone, (6.12) with $d = 2$, $s = 5$ holds on $[\beta_1, \beta_2]$ as soon as the upward-rounded vector $\Gamma^{\circ 5}(\tau^{(7)}(\beta_1))$ is bounded by the downward-rounded vector $\tau^{(2)}(\beta_2)$ at the indices $\ell \in \{1, 2, 3\}$, which we verified for every subinterval (`code/verify_uniform_bounds_and_plane.py`). Proposition 6.11 then gives $\theta^{\mathrm{D}}_7(\beta) > 0$. Hence $\beta^{\mathrm{D}}_c(7) \leq \beta^{\mathrm{D}}_c(2)$. □

Remark 6.16. For $\beta = 2$ the column projection is the reduction used in [LXZ26] to show that the directed two-neighbour graph percolates in every dimension $d \geq 3$, with $\theta_d^{\mathrm{D}}(2) \geq \theta_2^{\mathrm{bond}}(11/20)$. The obstruction to a direct comparison of consecutive dimensions is discussed in Section 8.

7 The planar bidirectional graph

In this section $d = 2$, and we consider the bidirectional rule $\mathrm{R} = \wedge$ only. We write $\mathbb{P}_\mu := \mathbb{P}_\mu^\wedge$, $\mathcal{D} := \mathcal{D}_2$, $\upsilon := \upsilon_{2,3} = Q_{2,3}$, $\pi_q := \pi_{2,q}$ and $q_0 := 1/\sqrt{2}$, so that $\Psi_\upsilon(\{u\},\{v\}) = 1/2 = \Psi_{\pi_{q_0}}(\{u\},\{v\})$ for $u \neq v$.

7.1 Pivotal vertices and a Russo formula

In the notation of Lemma 3.1 we call $a \in B_n$ *pivotal* if $T_o^* \neq \emptyset$ (for $a = o$), if C_a^{c} occurs and $O_a^* \neq \emptyset$ (for $a \in \partial B_n$), or if C_a^{c} occurs and $O_a^* \neq \emptyset \neq T_a^*$ (otherwise). We write Piv_a for this $\mathcal{G}_a$-measurable event and

$$\Xi_\mu(a) := \begin{cases} \Psi_\mu(T_o^*) & \text{if } a = o\,, \\ \Psi_\mu(O_a^*) & \text{if } a \in \partial B_n\,, \\ \Psi_\mu(T_a^*, O_a^*) & \text{otherwise}\,. \end{cases}$$

By Lemma 3.1, $\mathbb{P}_\mu(o \longleftrightarrow \partial B_n \,|\, \mathcal{G}_a) = \mathbb{1}_{C_a} + \mathbb{1}_{\mathrm{Piv}_a}\Xi_\mu(a)$, since $\mu(E_a) = 0$ on $C_a^{\mathrm{c}} \setminus \mathrm{Piv}_a$. The pair (T_a^*, O_a^*), or the single set T_o^* or O_a^*, is the *type* of a pivotal vertex.

Lemma 7.1. *Let $\lambda \mapsto \mu_\lambda$ be a family of local laws such that $\lambda \mapsto \mu_\lambda(S)$ is differentiable for every $S \subseteq \mathcal{D}$, and write $\partial_\lambda \Xi_{\mu_\lambda}(a)$ for the derivative of $\lambda \mapsto \Xi_{\mu_\lambda}(a)$ at a fixed type. Then, for every $n \in \mathbb{N}$,*

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbb{P}_{\mu_\lambda}(o \longleftrightarrow \partial B_n) = \sum_{a \in B_n} \mathbb{E}_{\mu_\lambda}\big[\mathbb{1}_{\mathrm{Piv}_a}\, \partial_\lambda \Xi_{\mu_\lambda}(a)\big]\,.$$

Proof. The probability is a finite sum of products $\prod_{x \in B_n} \mu_\lambda(S_x)$ over configurations $(S_x)_{x \in B_n}$ in $E := \{o \longleftrightarrow \partial B_n\}$. By the product rule, the term for a equals $\mathbb{E}_{\mu_\lambda}\big[\sum_S \partial_\lambda \mu_\lambda(S)\, \mathbb{1}_E\big((N_x)_{x \neq a}, S\big)\big]$, where $((N_x)_{x \neq a}, S)$ denotes the configuration with N_a replaced by S. By Lemma 3.1 the inner sum equals $\mathbb{1}_{C_a}\partial_\lambda \sum_S \mu_\lambda(S) + \mathbb{1}_{\mathrm{Piv}_a}\partial_\lambda \Xi_{\mu_\lambda}(a)$, and the first term vanishes because $\sum_S \mu_\lambda(S) = 1$. □

7.2 Strategy

For $q, t \in [0,1]$ let

$$\mu_{q,t} := (1-t)\,\pi_q + t\,\upsilon\,, \qquad f_n(q,t) := \mathbb{P}_{\mu_{q,t}}(o \longleftrightarrow \partial B_n)\,.$$

The function f_n is a polynomial in (q,t), and $f_n(q,0)$ is the corresponding probability for Bernoulli bond percolation with parameter q^2. The idea is to compare the effect of the two parameters on f_n. By Lemma 7.1, both partial derivatives are sums over pivotal vertices. Increasing t replaces the product law by the uniform law. Near $q = q_0$ this helps strictly at every pivotal vertex except at those of type $(\{u\},\{v\})$ (Lemma 7.2), at which it is neutral for $q = q_0$ by Lemma 3.4. We call these pivotal vertices *tight*. A local surgery shows that tight pivotal vertices are, up to a constant, not more likely than non-tight ones (Lemma 7.3). This yields a differential inequality $\partial_t f_n \geq \gamma\, \partial_q f_n$ near $(q_0, 0)$ with $\gamma > 0$ independent of n (Lemma 7.4). Integrating it along a line of slope $-\gamma$ as in [AG91] shows that a supercritical bond percolation is dominated by a model $\mu_{q',t}$ with $q' < q_0$ (Proposition 7.5). Finally, $Q_{2,3-\delta'}$ dominates $\mu_{q',t}$ in the sense of (3.3) for small $\delta' > 0$.

7.3 The local differences

For a pivotal vertex a let $\Delta_q(a) \coloneqq \Xi_\upsilon(a) - \Xi_{\pi_q}(a)$. A pivotal vertex $a \in B_n \setminus (\{o\} \cup \partial B_n)$ is *tight* if $|T_a^*| = |O_a^*| = 1$; all other pivotal vertices, including all pivotal vertices in $\{o\} \cup \partial B_n$, are *non-tight*. We write $\mathrm{Piv}_a^{\mathrm{t}}$ and $\mathrm{Piv}_a^{\mathrm{nt}}$ for the corresponding events, both $\mathcal{G}_a$-measurable.

Lemma 7.2. *Let $|q - q_0| \le \delta \le 1/1600$. On $\mathrm{Piv}_a^{\mathrm{nt}}$ we have $\Delta_q(a) \ge 1/400$, and on $\mathrm{Piv}_a^{\mathrm{t}}$ we have $\Delta_q(a) \ge -2\delta$. Moreover, $0 \le \partial_q \Xi_{\pi_q}(a) \le 4$ on Piv_a, and $\Delta_{q_0}(a) \ge 1/200$ on $\mathrm{Piv}_a^{\mathrm{nt}}$.*

Proof. For an increasing event of the four independent indicators of π_q, Russo's formula gives $0 \le \partial_q \pi_q(\cdot) \le 4$. This proves the third claim and shows that $q \mapsto \Xi_{\pi_q}(a)$ is 4-Lipschitz. Since υ is the law of $\mathcal{D} \setminus \{R\}$ with R uniform on $\mathcal{D}$, the non-tight types have the following values, where $i = |A|$ and $j = |B|$:

$$\Psi_\upsilon(A) = \begin{cases} 3/4 & i = 1\,, \\ 1 & i \ge 2\,, \end{cases} \qquad \Psi_\upsilon(A,B) = \begin{cases} 3/4 & \min\{i,j\} = 1 < \max\{i,j\}\,, \\ 1 & i = j = 2\,. \end{cases}$$

With $r \coloneqq 1 - q_0$, the corresponding values of π_{q_0} are $1 - r^i$ and $(1 - r^i)(1 - r^j)$. The differences Δ_{q_0} are $3/4 - q_0 \approx 0.0429$, $r^2 \approx 0.0858$, $r^3 \approx 0.0251$, $r^4 \approx 0.0074$ for the single sets, and $3/4 - q_0(1 - r^2) \approx 0.1036$, $3/4 - q_0(1 - r^3) \approx 0.0607$, $1 - (1 - r^2)^2 \approx 0.1642$ for the pairs; see `code/local_differences_planar.py`. All are at least $r^4 \ge 1/200$, and by the Lipschitz property $\Delta_q(a) \ge 1/200 - 4\delta \ge 1/400$ on $\mathrm{Piv}_a^{\mathrm{nt}}$. On $\mathrm{Piv}_a^{\mathrm{t}}$ we have $\Delta_q(a) = 1/2 - q^2 \ge 1/2 - (q_0 + \delta)^2 = -2q_0\delta - \delta^2 \ge -2\delta$. □

7.4 The surgery

For $a \in \mathbb{Z}^2$ let $\Lambda_a \coloneqq (a + \{-2, \dots, 2\}^2) \cap B_n$. Then $\#\Lambda_a \le 25$, and every vertex lies in at most 25 of the sets Λ_a.

Lemma 7.3. *Let $n \ge 5$, $q \in [3/5, 4/5]$, $t \in [0, 1/2]$ and $m_0 \coloneqq 1/1250$. Then for every $a \in B_n \setminus (\{o\} \cup \partial B_n)$,*

$$\mathbb{P}_{\mu_{q,t}}(\mathrm{Piv}_a) \le m_0^{-25} \sum_{a' \in \Lambda_a} \mathbb{P}_{\mu_{q,t}}(\mathrm{Piv}_{a'}^{\mathrm{nt}})\,.$$

Proof. Write $\Lambda \coloneqq \Lambda_a$, let $\omega_{\mathrm{out}} \coloneqq (N_x)_{x \in B_n \setminus \Lambda}$, and let G_{out} be the graph on $B_n \setminus \Lambda$ formed by the open edges with both endpoints outside Λ; it is a function of ω_{out}. Every set $S \subseteq \mathcal{D}$ satisfies $\mu_{q,t}(S) \ge (1 - t)\min\{q, 1 - q\}^4 \ge \frac{1}{2} \cdot 5^{-4} = m_0$.

Reduction. Let F be the set of outside configurations ω_{out} for which some choice of $(N_x)_{x \in \Lambda}$ makes a pivotal. Since Piv_a does not depend on N_a, $\mathrm{Piv}_a \subseteq \{\omega_{\mathrm{out}} \in F\}$. We show below that there is a map ξ, assigning to every $\omega_{\mathrm{out}} \in F$ a choice $\xi(\omega_{\mathrm{out}}) = (S_x)_{x \in \Lambda}$, such that the configuration $(\omega_{\mathrm{out}}, \xi(\omega_{\mathrm{out}}))$ lies in $\mathrm{Piv}_{a'}^{\mathrm{nt}}$ for some $a' \in \Lambda$. Then, by independence,

$$\sum_{a' \in \Lambda} \mathbb{P}(\mathrm{Piv}_{a'}^{\mathrm{nt}}) \ge \mathbb{P}\big(\omega_{\mathrm{out}} \in F,\ (N_x)_{x \in \Lambda} = \xi(\omega_{\mathrm{out}})\big) \ge m_0^{\#\Lambda}\, \mathbb{P}(\omega_{\mathrm{out}} \in F) \ge m_0^{25}\, \mathbb{P}(\mathrm{Piv}_a)\,.$$

Ports. Fix $\omega_{\mathrm{out}} \in F$ and a completion making a pivotal; we may set $N_a = \mathcal{D}$, since this does not affect Piv_a. Then $o \longleftrightarrow \partial B_n$ by Lemma 3.1, and every open path from o to ∂B_n in B_n visits a, since C_a fails. In particular, o and ∂B_n are not connected in G_{out}, a subgraph of the open graph without a. Let $\pi = (\pi_0, \dots, \pi_\ell)$ be a self-avoiding open path from o to ∂B_n in B_n that visits ∂B_n only at its end. It visits a, which is neither o nor in ∂B_n. We call a pair (x, y) with $x \in \Lambda$,

$y \in B_n \setminus \Lambda$, $|x-y|=1$ and $x-y \in N_y$ an *o-port* if y is connected to o in G_{out}, and a *∂-port* if y is connected to ∂B_n in G_{out}. These notions depend only on ω_{out}. If $o \notin \Lambda$, no pair is both, since o and ∂B_n are not connected in G_{out}, and the outside vertex of an o-port is not in ∂B_n. In each of the three cases below, $\xi(\omega_{\mathrm{out}})$ is defined from the lexicographically first ports with the required properties and from a deterministic construction, so that it depends on ω_{out} only. In every case all vertices of Λ outside an explicit subgraph H keep no edge, and the vertices of H keep exactly the edges listed. Then the open edges with at least one endpoint in Λ are exactly the listed edges.

Case 1: $3 \le \|a\|_\infty \le n-3$. Then $o \notin \Lambda$ and $\Lambda = a + \{-2,\dots,2\}^2 \subseteq B_{n-1}$. Let $i_1 := \min\{i\colon \pi_{i+1} \in \Lambda\}$ and $i_2 := \max\{i\colon \pi_{i-1} \in \Lambda\}$. The pieces $(\pi_0,\dots,\pi_{i_1})$ and $(\pi_{i_2},\dots,\pi_\ell)$ lie in G_{out}, so that (π_{i_1+1},π_{i_1}) is an o-port and (π_{i_2-1},π_{i_2}) is a ∂-port. Moreover $\pi_{i_1+1} \neq \pi_{i_2-1}$, since otherwise π would meet Λ in a single vertex, which would have to be a, while a has no neighbour outside Λ. Hence there are an o-port (x_1,y_1) and a ∂-port (x_2,y_2) with $x_1 \neq x_2$, and we take the lexicographically first such pair; both x_1, x_2 lie on the ring $R := \{x\colon \|x-a\|_\infty = 2\}$, which is a cycle of 16 vertices.

The gadget consists of the square a, $a+e_1$, $a+e_1+e_2$, $a+e_2$ and four spokes, see Figure 6: the two *a-spokes* $(a-2e_1, a-e_1)$ and $(a-2e_2, a-e_2)$, which end next to a, and the two *c-spokes* $(a+2e_1+e_2)$ and $(a+e_1+2e_2)$, which end next to $c := a+e_1+e_2$. The spokes are pairwise disjoint, disjoint from the square, and each meets R exactly in its first vertex, its *end*. In the cyclic order of R the ends appear as $a+2e_1+e_2$, $a+e_1+2e_2$, $a-2e_1$, $a-2e_2$, so that the two ends of each kind are adjacent in this order.

We claim that there are disjoint paths P_1 from x_1 to an end s_1 and P_2 from x_2 to an end s_2, both within R, such that s_1 and s_2 are of different kinds. Cut R at x_1 and x_2 into two arcs ρ, ρ', each containing x_1 and x_2. If one arc, say ρ, contains ends of both kinds, let s_1 be the first end on ρ seen from x_1 and s_2 an end of the other kind on ρ; then s_2 comes after s_1, and the sub-arcs from x_1 to s_1 and from s_2 to x_2 are disjoint. Otherwise each arc contains ends of one kind only, and since all four ends lie in $\rho \cup \rho'$, both arcs contain ends, of different kinds. Let P_1 run along ρ from x_1 to its first end and P_2 along ρ' from x_2 to its first end. These paths are disjoint: they could only meet in x_1 or x_2, which would then be an end lying on both arcs, contradicting that the arcs carry different kinds.

Let H be the graph whose edges are the edges between consecutive vertices of P_1, of P_2 and of the spokes at s_1 and s_2, the edge between the last vertex of the a-spoke and a, the edge between the last vertex of the c-spoke and c, and the four edges of the square. Every vertex of H keeps exactly its edges in H; in addition x_1 keeps $\{x_1,y_1\}$ and x_2 keeps $\{x_2,y_2\}$. All other vertices of Λ keep nothing. Then the open edges meeting Λ are the edges of H together with $\{x_1,y_1\}$ and $\{x_2,y_2\}$: they form a path from y_1 through P_1 and the spoke at s_1 to the square, the square, and a path from the square through the spoke at s_2 and P_2 to y_2.

Consider the new configuration and $a' := a$. Removing a splits the open edges meeting Λ into a piece attached to y_1 and a piece attached to y_2. The piece attached to y_1 contains no vertex of ∂B_n, because $\Lambda \cap \partial B_n = \emptyset$, and in G_{out} the cluster of o contains y_1 but no vertex of ∂B_n and not y_2. Hence C_a fails. If s_1 is an a-end, then O_a^* consists of the direction of the last vertex of the spoke at s_1, and $T_a^* = \{e_1,e_2\}$; if s_1 is a c-end, then $O_a^* = \{e_1,e_2\}$ and T_a^* consists of the direction of the last vertex of the spoke at s_2. The remaining neighbour of a keeps nothing. In both cases a is pivotal with $\{|T_a^*|, |O_a^*|\} = \{1,2\}$, hence non-tight.

Case 2: $\|a\|_\infty \le 2$. Then $o \in \Lambda$, $\Lambda \subseteq B_4$, and $\Lambda \cap \partial B_n = \emptyset$ since $n \ge 5$. With i_2 as in Case 1 there is a ∂-port, and we take the lexicographically first one, (x_2,y_2). Let H be a shortest path in the box Λ from o to x_2; its vertices keep the edges along H, and x_2 also keeps $\{x_2,y_2\}$. Then the neighbour of o on $H \cup \{y_2\}$ keeps the edge to o and is connected to ∂B_n without using o, so that

$T_o^* \neq \emptyset$, and the origin is pivotal, which is non-tight by definition.

Case 3: $n-2 \le \|a\|_\infty \le n-1$. Then $o \notin \Lambda$ because $n \ge 5$, and Λ is a box meeting ∂B_n. With i_1 as in Case 1 we see that an o-port exists, and we take the lexicographically first one, (x_1, y_1); then $y_1 \notin \partial B_n$. Let H be a shortest path in Λ from x_1 to $\Lambda \cap \partial B_n$, ending at z; only its last vertex lies in ∂B_n, and $H = \{x_1\}$ if $x_1 \in \partial B_n$. The vertices of H keep the edges along H, and x_1 also keeps $\{x_1, y_1\}$. Removing z, the open edges meeting Λ form a path from y_1 that contains no vertex of ∂B_n, and the cluster of o in G_{out} contains no vertex of ∂B_n. Hence C_z fails, the neighbour of z on $H \cup \{y_1\}$ gives a direction in O_z^*, and $z \in \partial B_n$ is pivotal, hence non-tight.

The three cases cover all $a \in B_n \setminus (\{o\} \cup \partial B_n)$, since $n - 2 \ge 3$. □

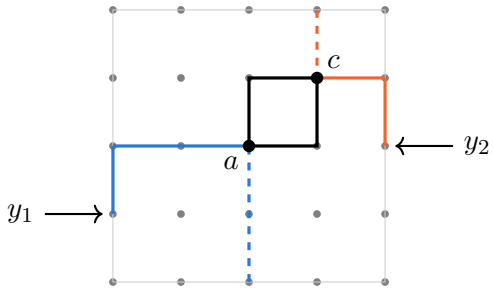


Figure 6: The surgery of Case 1 in the proof of Lemma 7.3 in the box $a + \{-2, \dots, 2\}^2$. The a-spokes are drawn in blue, the c-spokes in orange; one spoke of each kind (solid) is used. Here $x_1 = a - 2e_1 - e_2$ is connected to the a-end $a - 2e_1$ and $x_2 = a + 2e_1$ to the c-end $a + 2e_1 + e_2$ along the ring. After the surgery a is pivotal with $|T_a^*| = 2$ and $|O_a^*| = 1$.

7.5 The differential inequality and the proof of Theorem 2.6

Let $K := 25\, m_0^{-25}$, $\delta := 1/(1600K)$ and $\gamma := 1/(3200(1+K))$.

Lemma 7.4. *For all $n \ge 5$, $q \in [q_0 - \delta, q_0 + \delta]$ and $t \in [0, 1/2]$,*

$$\partial_t f_n(q,t) \ge \gamma\, \partial_q f_n(q,t) \ge 0\,.$$

Proof. Write $\mathbb{P} = \mathbb{P}_{\mu_{q,t}}$ and $S_{\text{t}} := \sum_{a \in B_n} \mathbb{P}(\text{Piv}_a^{\text{t}})$, $S_{\text{nt}} := \sum_{a \in B_n} \mathbb{P}(\text{Piv}_a^{\text{nt}})$. Since $\partial_t \mu_{q,t} = \upsilon - \pi_q$ and $\partial_q \mu_{q,t} = (1-t)\partial_q \pi_q$, Lemma 7.1 and Lemma 7.2 give

$$\partial_t f_n = \sum_{a \in B_n} \mathbb{E}\big[\mathbb{1}_{\text{Piv}_a} \Delta_q(a)\big] \ge \frac{S_{\text{nt}}}{400} - 2\delta S_{\text{t}}\,, \qquad 0 \le \partial_q f_n \le 4\,(S_{\text{t}} + S_{\text{nt}})\,.$$

Since $q \in [3/5, 4/5]$, Lemma 7.3 applies, and since every a' lies in at most 25 of the sets Λ_a, it gives $S_{\text{t}} \le K S_{\text{nt}}$. Hence $\partial_t f_n \ge (1/400 - 2\delta K) S_{\text{nt}} = S_{\text{nt}}/800$ and $\partial_q f_n \le 4(1+K) S_{\text{nt}}$, which gives the claim. □

Proposition 7.5. *Let $t := 1/2$. There is $q' < q_0$ such that the bidirectional model with local law $\mu_{q',t}$ percolates.*

Proof. Let $q_1 := q_0 + \gamma/4$, $q' := q_0 - \gamma/4$ and $g_n(\lambda) := f_n(q_1 - \gamma\lambda, \lambda)$ for $\lambda \in [0, 1/2]$. Along this segment $|q_1 - \gamma\lambda - q_0| \le \gamma/4 \le \delta$, so that Lemma 7.4 gives $g_n'(\lambda) = \partial_t f_n - \gamma\, \partial_q f_n \ge 0$ for $n \ge 5$. Hence

$$f_n(q', t) = g_n(t) \ge g_n(0) = f_n(q_1, 0) = \mathbb{P}_{q_1^2}^{\text{bond}}(o \longleftrightarrow \partial B_n) \ge \theta_2^{\text{bond}}(q_1^2)$$

for all $n \geq 5$, where θ_2^{bond} is the percolation probability of Bernoulli bond percolation on $\mathbb{Z}^2$. Since $q_1^2 > 1/2$, the right-hand side is positive by [Kes80], and letting $n \to \infty$ gives the claim. □

Proof of Theorem 2.6. The lower bound is Table 2. For the upper bound let $t = 1/2$ and q' be as in Proposition 7.5. We show that $Q_{2,3-\delta'} = (1-\delta')\upsilon + \delta'\upsilon_{2,2}$ satisfies (3.3) with respect to $\mu_{q',t}$ for small $\delta' > 0$. For every type T (a pair (A, B) or a single set A),

$$\Psi_{Q_{2,3-\delta'}}(T) - \Psi_{\mu_{q',t}}(T) = (1-t)\big(\Psi_\upsilon(T) - \Psi_{\pi_{q'}}(T)\big) - \delta'\big(\Psi_\upsilon(T) - \Psi_{\upsilon_{2,2}}(T)\big)\,.$$

The first term is at least $\eta := (1-t)\min\{1/2 - q'^2, 1/200\} > 0$: for $|A| = |B| = 1$ it equals $(1-t)(1/2-q'^2)$, and for all other types $\Psi_\upsilon - \Psi_{\pi_{q'}} \geq \Psi_\upsilon - \Psi_{\pi_{q_0}} \geq 1/200$ by Lemma 7.2. The second term is at most δ'. Hence (3.3) holds for $\delta' \leq \eta$, and Proposition 3.2 together with Proposition 7.5 gives $\theta_2^{\mathrm{B}}(3-\delta') > 0$. □

8 Discussion

The undirected graph. The rigorous upper bound $\beta_{\mathrm{c}}^{\mathrm{U}}(d) \leq \beta_{\mathrm{c}}^{\mathrm{D}}(d) \leq 1 + 1/(2d) + O(d^{-2})$ is far from the simulated values, while the lower bound of Theorem 2.5 is close to them, see Table 1.

Conjecture 8.1. *$\beta_{\mathrm{c}}^{\mathrm{U}}(d) = 1 + (1+o(1))/(8d^2)$ as $d \to \infty$.*

The lower bound comes from counting self-avoiding paths, which is asymptotically sharp for many high-dimensional models. A matching upper bound requires a comparison that does not pass through the directed graph. The comparison of Theorem 2.2 with Bernoulli bond percolation of parameter $1-(1-q)^2 \approx 2q$ only gives $\beta_{\mathrm{c}}^{\mathrm{U}}(d) \leq 1 + 1/8 + o(1)$, since it loses the information that every vertex keeps at least one edge. At $\beta = 1$ the undirected graph is the undirected one-neighbour graph, which does not percolate [JKLT25, Proposition 2.5], and Theorem 2.5 shows that a positive density of vertices with two kept edges is needed in every dimension. The strict inequality $\beta_{\mathrm{c}}^{\mathrm{U}}(d) < \beta_{\mathrm{c}}^{\mathrm{D}}(d)$, which the simulations show with a large margin, is open in every dimension.

The bidirectional graph. For $d \geq 11$ the bounds of Table 2 determine $\beta_{\mathrm{c}}^{\mathrm{B}}(d)$ within 0.03, and for $3 \leq d \leq 10$ the upper bounds are limited by the available rigorous bounds on $p_{\mathrm{c}}(d)$. The simulations suggest $\beta_{\mathrm{c}}^{\mathrm{B}}(d) - \varphi^{-1}(2d) \approx 0.53\, d^{-3/2}$, in line with the order of the error term in Theorem 2.5. For $2 \leq d \leq 8$ the simulated values lie strictly below the comparison value $\varphi^{-1}(2d(2d-1)p_{\mathrm{c}}(d))$ evaluated with the numerical estimates of $p_{\mathrm{c}}(d)$ of [MM18], by margins between 0.16 ($d = 2$) and 0.009 ($d = 8$); for instance $\beta_{\mathrm{c}}^{\mathrm{B}}(6) \approx 4.036 < 4.054$. For $d \in \{9, 10\}$ the margins are within the statistical errors. As for $d = 2$ (Theorem 2.6), we expect that an enhancement argument gives this strict inequality in general.

The directed graph. Theorem 2.5 determines $\beta_{\mathrm{c}}^{\mathrm{D}}(d)$ up to $O(d^{-2})$. More precisely, $c(d) < 2d - 1$ and the expansion $p_{\mathrm{c}}(d) = (2d)^{-1} + (2d)^{-2} + \frac{7}{2}(2d)^{-3} + O(d^{-4})$ of [HS95, vdHS06] give $\beta_{\mathrm{c}}^{\mathrm{D}}(d) = 1 + (2d)^{-1} + a_d(2d)^{-2}$ with $1 + o(1) \leq a_d \leq \frac{7}{2} + o(1)$. The bounds of Table 3 decrease in d for $d \leq 30$, and the simulations suggest the following, in line with [JKLT25, Conjecture 3.2].

Conjecture 8.2. *The map $d \mapsto \beta_{\mathrm{c}}^{\mathrm{D}}(d)$ is strictly decreasing.*

In contrast to $p_{\mathrm{c}}(d)$, which decreases because $\mathbb{Z}^d$ embeds into $\mathbb{Z}^{d+1}$, the mean out-degree is fixed when the dimension grows, so that the additional neighbours dilute the choices, and no inclusion argument is available. For Bernoulli bond percolation, the expansion of $p_{\mathrm{c}}(d)$ in [HS95, vdHS06] shows that $d \mapsto 2d\, p_{\mathrm{c}}(d)$ is strictly decreasing for all sufficiently large d, with a threshold that is not

explicit; as far as we know, monotonicity in low dimensions, including the first step $4p_c(2) > 6p_c(3)$, is open there as well. If (6.12) held with $s = 1$ for all β in a neighbourhood of $\beta_c^D(d)$, then Proposition 6.11 would give $\beta_c^D(d+1) \le \beta_c^D(d)$, that is, the non-strict version of Conjecture 8.2. For $d \in \{2, 3\}$ the set of $\beta \in [1, 2]$ for which (6.12) holds with $s = 1$ is empty: at $\ell = 2d - 1$ the difference of the two sides is a rational function of β without zeros in $[1, 2]$ and of the wrong sign. For larger d we computed these sets on a grid, with endpoints accurate to within 0.005: $[1.69, 1.89]$ for $d = 4$, $[1.59, 1.96]$ for $d = 6$ and $[1.55, 1.99]$ for $d = 10$ (`code/obstruction_intervals.py`). These intervals lie above the simulated critical values. With more projection steps they grow: for $d = 2$ they are approximately $[1.62, 1.91]$, $[1.52, 1.96]$ and $[1.45, 1.98]$ for $s = 3, 4, 5$, and $[1.30, 2.00]$ for $s = 10$. The difference $\tau_\ell^{(d)}(\beta) - \Gamma(\tau^{(d+1)}(\beta))_\ell$ is positive for small ℓ, since the column collects on average more than β horizontal directions, and negative for ℓ close to $2d$, because the projected out-set is empty or a single direction with a probability that a degree-constrained law cannot match; for instance, for $d = 2$ and $\beta = 1.84$ the differences are 0.042, 0.020, -0.009 for $\ell = 1, 2, 3$.

We see four routes towards Conjecture 8.2. First, Proposition 6.11 with $s = 4$ gives $\beta_c^D(6) \le \beta_c^D(2)$ as soon as $\beta_c^D(2) \ge 1.52$ is known rigorously, and any rigorous improvement of (2.8) translates into new comparisons. Second, interpolating between $Q_{d,\beta}$ and the projected law site by site, the derivative of the one-arm probability is a sum over pivotal vertices of the local differences above; the negative contributions come from pivotal vertices whose set of good directions is almost all of $\mathcal{D}_d$, and a local modification of a neighbour could trade them against the gains at small ℓ. Third, richer explorations than single columns, for instance of whole fibres, collect more choices, but an exactly computable version is needed. Fourth, for $\beta = 2$ Lemma 6.13 holds for all $d \ge 2$, and whether $\theta_d^D(2)$ itself is non-decreasing in d is open.

Monotonicity in the dimension. The simulations suggest that β_c^U and β_c^D are decreasing and β_c^B is increasing in d, and that $\beta_c^B(d) - \varphi^{-1}(2d)$ is decreasing. None of these statements is proved. The intervals of Table 2 for $d \in \{11, 12, 15, 20\}$ are disjoint and increasing, so that at least $\beta_c^B(11) < \beta_c^B(12) < \beta_c^B(15) < \beta_c^B(20)$.

Planar values. In the plane, $\beta_c^B(2) \approx 2.840$, $\beta_c^D(2) \approx 1.84$ and $\beta_c^U(2) \approx 1.249$. We are not aware of an exact solution or a duality relation for any of them. Since the discarded sets $\mathcal{D}_2 \setminus N_x$ have law $Q_{2,4-\beta}$, the closed edges of the undirected graph form a bidirectional graph of the discarded sets, but on the primal rather than the dual lattice, so that planar duality does not apply directly.

Acknowledgements

The author acknowledges the financial support of the Leibniz Association within the Leibniz Junior Research Group on Probabilistic Methods for Dynamic Communication Networks (DYCOMNET) as part of the Leibniz Competition (grant no. J105/2020), of the Berlin Mathematics Research Center MATH+ through the project EF-MA-Sys-2 on Information Flow & Emergent Behavior in Complex Networks, and of the Deutsche Forschungsgemeinschaft (DFG) through Project P27 within the SPP 2265.

Use of AI tools

The simulations, proofs, computations and the text of this note were developed with substantial assistance of the AI system Claude (Anthropic), in an interactive process directed by the author.

Code

The code is available from the author on request and in the GitHub repository `https://github.com/jonaskoeppl/percolation-discrete-knn`. The simulations use the C programs `sim_spanning_thresholds.c` (spanning thresholds in boxes) and `sim_mean_cluster_size.c` (mean cluster size on the infinite lattice), with `run_spanning_simulations.sh` as driver; the raw output is in `simulation_data.tar.gz`. The Python scripts are the following.

- `estimate_from_spanning.py`: crossing points of the spanning-threshold distributions with bootstrap errors ($d \leq 4$).
- `cluster_size_fit_bidirectional.py`, `cluster_size_fit_undirected.py`, `cluster_size_fit_directed.py`: adaptive choice of β, mean cluster sizes and extrapolation of $\chi^{-1/\gamma}$ for the three graphs ($d \geq 3$).
- `rigorous_bounds.py`: the bounds of Table 2, from the published bounds on $p_c(d)$ and $c(d)$, in exact rational arithmetic.
- `make_tables_and_figures.py`: generates Tables 1 and 2 and Figures 2 and 3.
- `make_model_figure.py`: generates Figure 1.
- `check_mixture_inequality.py`: numerical check of Lemma 3.4, see Remark 3.7.
- `local_differences_planar.py`: the local differences in Lemma 7.2.
- `verify_uniform_bounds_and_plane.py`: certification of Theorems 2.7 and 2.9 in exact rational arithmetic.
- `certify_consecutive_dimensions.py`: the identity (6.14) and the Bernstein certificate in the proof of Lemma 6.13.
- `obstruction_intervals.py`: the intervals and differences in the discussion of Conjecture 8.2.

References

[AG91] M. Aizenman and G. Grimmett, *Strict monotonicity for critical points in percolation and ferromagnetic models*, J. Stat. Phys. **63** (1991), no. 5-6, 817–835.

[BJK+25] J. Bäumler, B. Jahnel, J. Köppl, B. Lodewijks, L. Reeves, and A. Tóbiás, *Local criteria for global connectivity comparisons: beyond stochastic domination*, 2025, arXiv:2510.03934.

[CHJK26] D. Coupier, B. Henry, B. Jahnel, and J. Köppl, *The planar lattice two-neighbor graph percolates*, Ann. Probab. (2026), To appear. arXiv:2412.20781.

[FvdH17] R. Fitzner and R. van der Hofstad, *Mean-field behavior for nearest-neighbor percolation in $d > 10$*, Electron. J. Probab. **22** (2017), Paper No. 43, 1–65.

[GPS26] P. A. Gomes, A. Pereira, and R. Sanchis, *Upper bounds for critical probabilities in Bernoulli percolation models*, J. Appl. Probab. **63** (2026), no. 1, 61–72.

[HS95] T. Hara and G. Slade, *The self-avoiding-walk and percolation critical points in high dimensions*, Combin. Probab. Comput. **4** (1995), no. 3, 197–215.

[JKLT25] B. Jahnel, J. Köppl, B. Lodewijks, and A. Tóbiás, *Percolation in lattice k-neighbor graphs*, J. Appl. Probab. **62** (2025), no. 4, 1475–1492.

[Kes80] H. Kesten, *The critical probability of bond percolation on the square lattice equals* 1/2, Comm. Math. Phys. **74** (1980), no. 1, 41–59.

[LXZ26] J. Liu, K. Xiang, and L. Zou, *Two-neighbor lattice percolation percolates in all dimensions*, 2026, arXiv:2610.38200.

[MM18] S. Mertens and C. Moore, *Percolation thresholds and Fisher exponents in hypercubic lattices*, Phys. Rev. E **98** (2018), no. 2, 022120.

[PT00] A. Pönitz and P. Tittmann, *Improved upper bounds for self-avoiding walks in $\mathbb{Z}^d$*, Electron. J. Combin. **7** (2000), Research Paper 21, 10 pp.

[vdHS06] R. van der Hofstad and G. Slade, *Expansion in n^{-1} for percolation critical values on the n-cube and $\mathbb{F}^n$: the first three terms*, Combin. Probab. Comput. **15** (2006), no. 5, 695–713.